A PROJECT REPORTED ON

CELESTIAL PATH EXPLORATION THROUGH A MULTIFACETED SIMULATION SYSTEM FOR TRAJECTORY ANALYSIS OF CELESTIAL BODIES

ABSTRACT:

The Project “Celestial Path Exploration through a Multifaceted Simulation System for Trajectory Analysis of Celestial Bodies” integrated system is designed to revolutionize energy data processing and grid model construction by incorporating the MLP Regressor. Its user-friendly interface caters to diverse user roles, allowing easy registration and submission of data sources. Ensure transparency in evaluating submissions, facilitating effective communication. Team members contribute to the system by uploading energy data sources based on requirements, establishing a robust foundation for grid model construction. The integration of power source and solid mass data along with the Cat Boost Regressor Algorithm provides crucial insights for precise modeling, while the incorporation of Solid Gravity Energy data along with Random Forest Regressor enhances system maintenance practices. With exclusive access, orchestrate the entire process, making informed decisions, notifying, and overseeing billing based on assessed results.. In addition to its efficiency, the system emphasizes sustainability through enabling users to identify consumption patterns and optimize resource usage. This user-centric design enhances overall energy infrastructure management, fostering effectiveness and user satisfaction. Furthermore, the system's adaptability to emerging technologies ensures its relevance in the face of industry advancements, providing a future-proof solution for evolving energy landscapes. Its proactive sustainability features position it not only as a tool for efficient energy management but also as a catalyst for positive change, promoting environmental responsibility and technological progress.

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CHAPTER-1

1.1 INTRODUCTION TO PROJECT

The project "Celestial Path Exploration through a Multifaceted Simulation System for Trajectory Analysis of Celestial Bodies" is a groundbreaking initiative designed to revolutionize energy data processing and grid model construction. By integrating a user-friendly interface, the system caters to diverse user roles, ensuring easy registration and submission of data sources. Transparency is a key focus, with the system facilitating effective communication and evaluation of submissions. Team members contribute to the system by uploading energy data sources, which are then used to establish a robust foundation for grid model construction. The integration of power source and solid mass data provides crucial insights for precise modeling, while the incorporation of Solid Gravity Energy data enhances system maintenance practices. The project emphasizes sustainability by enabling users to identify consumption patterns and optimize resource usage. Its user-centric design enhances overall energy infrastructure management, fostering effectiveness and user satisfaction. The system's adaptability to emerging technologies ensures its relevance in the face of industry advancements, providing a future-proof solution for evolving energy landscapes. Overall, the project not only revolutionizes energy data processing and grid model construction but also promotes environmental responsibility and technological progress in energy infrastructure management.

1.2 PURPOSE OF THE SYSTEM

- ✓ Revolutionize energy data processing and grid model construction.
- ✓ Provide a user-friendly interface for diverse user roles.
- ✓ Ensure transparency in evaluating submissions and facilitate effective communication.
- ✓ Establish a robust foundation for grid model construction through team collaboration in uploading energy data sources.
- ✓ Enable users to identify consumption patterns and optimize resource usage for sustainability.
- ✓ Adapt to emerging technologies to remain relevant in evolving energy landscapes.

CHAPTER-2

SYSTEM ANALYSIS

2.1 INTRODUCTION

The project "Celestial Path Exploration through a Multifaceted Simulation System for Trajectory Analysis of Celestial Bodies" is a groundbreaking initiative designed to revolutionize energy data processing and grid model construction. By integrating a user-friendly interface, the system caters to diverse user roles, ensuring easy registration and submission of data sources. Transparency is a key focus, with the system facilitating effective communication and evaluation of submissions. Team members contribute to the system by uploading energy data sources, which are then used to establish a robust foundation for grid model construction. The integration of power source and solid mass data provides crucial insights for precise modeling, while the incorporation of Solid Gravity Energy data enhances system maintenance practices. The project emphasizes sustainability by enabling users to identify consumption patterns and optimize resource usage. Its user-centric design enhances overall energy infrastructure management, fostering effectiveness and user satisfaction. The system's adaptability to emerging technologies ensures its relevance in the face of industry advancements, providing a future-proof solution for evolving energy landscapes. Overall, the project not only revolutionizes energy data processing and grid model construction but also promotes environmental responsibility and technological progress in energy infrastructure management.

2.2 ANALYSIS

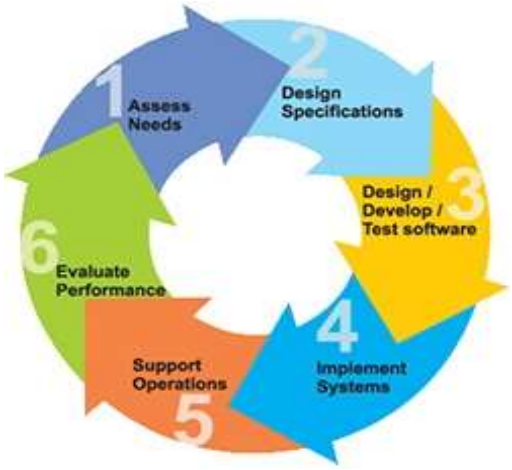
SOFTWARE DEVELOPMENT LIFE CYCLE

INTRODUCTION:

The System Development Lifecycle framework is designed to outline a complete development and implementation process suitable for developing complex applications. SDLC is a process followed for a software project, within a software organization. It consists of a detailed plan describing how to develop, maintain, replace and alter or enhance specific software. The life cycle defines a methodology for improving the quality of software and the overall development process.

- Business – legislation regulatory requirements, policy, SOP's, guidelines etc.
- Process – how the business is implemented
- Data – the core business data elements collected for the business
- Application – the gate to the business collecting
- Infrastructure- the servers, network, workstations, etc.

2.3 SDLC Phases:



Stage 1: Scheduling and Requisite investigation:

Requirement analysis is the most important and fundamental stage in SDLC. It is performed by the senior members of the team with inputs from the customer, the sales department, market surveys and domain experts in the industry. This information is then used to plan the basic project approach and to conduct product feasibility study in the economical, operational, and technical areas.

Planning for the quality assurance requirements and identification of the risks associated with the project is also done in the planning stage. The outcome of the technical feasibility study is to define the various technical approaches that can be followed to implement the project successfully with minimum risks.

Stage 2: Significant necessities:

Once the requirement analysis is done the next step is to clearly define and document the product requirements and get them approved from the customer or the market analysts. This is done through .SRS. . Software Requirement Specification document which consists of all the product requirements to be designed and developed during the project life cycle.

Stage 3: Scheming the product design:

SRS is the reference for product architects to come out with the best architecture for the product to be developed. Based on the requirements specified in the SRS, usually more than one design approach for the product architecture is proposed and documented in a DDS - Design Document Specification.

This DDS is reviewed by all the important stakeholders and based on various parameters as risk assessment, product robustness, design modularity, budget and time constraints, the best design approach is selected for the product.

Stage 4: Structure or Mounting the Product:

In this stage of SDLC the actual development starts and the product are built. The programming code is generated as per DDS during this stage. If the design is performed in a detailed and organized manner, code generation can be accomplished without much hassle.

Developers have to follow the coding guidelines defined by their organization and programming tools like compilers, interpreters, debuggers etc.are used to generate the code. Different high level programming languages such as C, C++, Pascal, Java, and PHP are used for coding.

Stage 5: Testing the Product:

This stage is usually a subset of all the stages as in the modern SDLC models, the testing activities are mostly involved in all the stages of SDLC. However, this stage refers to the testing only stage of the product, where product defects are reported, tracked, fixed and retested, until the product reaches the quality standards defined in the SRS.

Stage 6: Consumption in the Market and Safeguarding:

Once the product is tested and ready to be deployed it is released formally in the appropriate market. Sometime product deployment happens in stages as per the organizations. Business strategy. The product may first be released in a limited segment and tested in the real business environment (UAT- User acceptance testing).

The product may be released as it is or with suggested enhancements in the targeting market segment. After the product is released in the market, its maintenance is done for the existing customer base.

2.4 HARDWARE AND SOFTWARE REQUIREMENTS

Developing Kit			
	Processor	**RAM**	**Disk Space**
PyCharm	**Computer with a 2.6GHz processor or higher**	**2GB**	**Minimum 20 GB**
Database			
MySQL 5.0	**Intel Pentium processor at 2.6GHz or faster**	**Minimum 512 MB Physical Memory; 1 GB Recommended**	**Minimum 20 GB**

Software Requirements:

- **Front end** : HTML, CSS, JS
- **Web application** : PycharmIDE
- **Back end** : Core Python, Django Framework
- **Database** : MySQL5.6.12-log

OVERVIEW OF SOFTWARE ENGINEERING:

Software is more than just a program code. A program is an executable code, which serves some computational purpose. Software is considered to be collection of executable programming code, associated libraries and documentations. Software, when made for a specific requirement is called **software product.**

Engineering on the other hand, is all about developing products, using well-defined, scientific principles and methods.

Software engineering is an engineering branch associated with development of software product using well-defined scientific principles, methods and procedures. The outcome of software engineering is an efficient and reliable software product.

Definitions

IEEE defines software engineering as:

(1) The application of a systematic,disciplined,quantifiable approach to the development,operation and maintenance of software; that is, the application of engineering to software.

(2) The study of approaches as in the above statement.

Fritz Bauer, a German computer scientist, defines software engineering as:

Software engineering is the establishment and use of sound engineering principles in order to obtain economically software that is reliable and work efficiently on real machines.

Software Evolution

The process of developing a software product using software engineering principles and methods is referred to as **software evolution.** This includes the initial development of software and its maintenance and updates, till desired software product is developed, which satisfies the expected requirements.

Evolution starts from the requirement gathering process. After which developers create a prototype of the intended software and show it to the users to get their feedback at the early stage of software product development. The users suggest changes, on which several consecutive updates and maintenance keep on changing too. This process changes to the original software, till the desired software is accomplished.

Even after the user has desired software in hand, the advancing technology and the changing requirements force the software product to change accordingly. Re-creating software from scratch and to go one-on-one with requirement is not feasible. The only feasible and economical solution is to update the existing software so that it matches the latest requirements.

Software Evolution Laws

Lehman has given laws for software evolution. He divided the software into three different categories:

- **S-type (static-type) -** This is a software, which works strictly according to defined specifications and solutions. The solution and the method to achieve it, both are immediately understood before coding. The s-type software is least subjected to changes hence this is the simplest of all. For example, calculator program for mathematical computation.
- **P-type (practical-type) -** This is a software with a collection of procedures.This is defined by exactly what procedures can do. In this software, the specifications can be described but the solution is not obvious instantly. For example, gaming software.
- **E-type (embedded-type) -** This software works closely as the requirement of real-world environment. This software has a high degree of evolution as there are various changes in laws, taxes etc. in the real world situations. For example, Online trading software.

E-Type software evolution

Lehman has given eight laws for E-Type software evolution -

- **Continuing change -** An E-type software system must continue to adapt to the real world changes, else it becomes progressively less useful.
- **Increasing complexity -** As an E-type software system evolves, its complexity tends to increase unless work is done to maintain or reduce it.

- **Conservation of familiarity -** The familiarity with the software or the knowledge about how it was developed, why was it developed in that particular manner etc. must be retained at any cost, to implement the changes in the system.
- **Continuing growth-** In order for an E-type system intended to resolve some business problem, its size of implementing the changes grows according to the lifestyle changes of the business.
- **Reducing quality -** An E-type software system declines in quality unless rigorously maintained and adapted to a changing operational environment.
- **Feedback systems-** The E-type software systems constitute multi-loop, multi-level feedback systems and must be treated as such to be successfully modified or improved.
- **Self-regulation -** E-type system evolution processes are self-regulating with the distribution of product and process measures close to normal.
- **Organizational stability -** The average effective global activity rate in an evolving E-type system is invariant over the lifetime of the product.

Software Paradigms

Software paradigms refer to the methods and steps, which are taken while designing the software. There are many methods proposed and are in work today, but we need to see where in the software engineering these paradigms stand. These can be combined into various categories, though each of them is contained in one another:

Programming paradigm is a subset of Software design paradigm which is further a subset of Software development paradigm.

Software Development Paradigm

This Paradigm is known as software engineering paradigms where all the engineering concepts pertaining to the development of software are applied. It includes various researches and requirement gathering which helps the software product to build. It consists of –

- Requirement gathering
- Software design
- Programming

Software Design Paradigm

This paradigm is a part of Software Development and includes –

- Design
- Maintenance
- Programming

Programming Paradigm

This paradigm is related closely to programming aspect of software development. This includes –

- Coding
- Testing
- Integration

Need of Software Engineering

The need of software engineering arises because of higher rate of change in user requirements and environment on which the software is working.

- **Large software -** It is easier to build a wall than to a house or building, likewise, as the size of software become large engineering has to step to give it a scientific process.

- **Scalability-** If the software process were not based on scientific and engineering concepts, it would be easier to re-create new software than to scale an existing one.
- **Cost-** As hardware industry has shown its skills and huge manufacturing has lower down the price of computer and electronic hardware. But the cost of software remains high if proper process is not adapted.
- **Dynamic Nature-** The always growing and adapting nature of software hugely depends upon the environment in which user works. If the nature of software is always changing, new enhancements need to be done in the existing one. This is where software engineering plays a good role.
- **Quality Management-** Better process of software development provides better and quality software product.

Characteristics of good software

A software product can be judged by what it offers and how well it can be used. This software must satisfy on the following grounds:

- Operational
- Transitional
- Maintenance

Well-engineered and crafted software is expected to have the following characteristics:

Operational

This tells us how well software works in operations. It can be measured on:

- Budget
- Usability
- Efficiency
- Correctness
- Functionality
- Dependability

- Security
- Safety

Transitional

This aspect is important when the software is moved from one platform to another:

- Portability
- Interoperability
- Reusability
- Adaptability

Maintenance

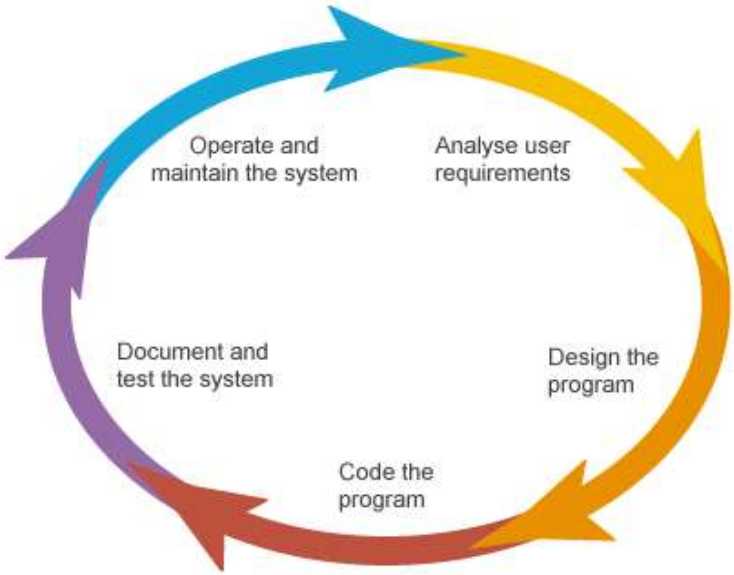
This aspect briefs about how well a software has the capabilities to maintain itself in the ever-changing environment:

- Modularity
- Maintainability
- Flexibility
- Scalability

In short, Software engineering is a branch of computer science, which uses well-defined engineering concepts required to produce efficient, durable, scalable, in-budget and on-time software products.

SOFTWARE DEVELOPMENT LIFE CYCLE

The Software Development Life Cycle is a process that ensures good software is built. Each phase in the life cycle has its own process and deliverables that feed into the next phase. There are typically 5 phases starting with the analysis and requirements gathering and ending with the implementation. Let's look in greater detail at each phase:



Stage 1: Scheduling and Requisite Analysis

During the discovery phase our team conducts a detailed requirement analysis and creates a work-breakdown structure.

Stage 2: Scheming the product design

We identify the design and architecture of the project. SRS is the reference for product architects to come out with the best architecture for the product to be developed.

Stage 3: Structure or Mounting the Product

In this stage of SDLC the actual development starts and the product is built. Different high level programming languages such as C, C++, Pascal, Java, C# and PHP are used for coding.

Stage 4: Testing the Product

Testing is the last phase of the Software Development Life Cycle before the software is delivered to customers. During testing, experienced testers start to test the system against the requirements.

Stage 5: Consumption in the Market and Safeguarding

Once the product has been fully tested and no high priority issues remain in the software, it is time to deploy to production where customers can use the system.

2.5 INPUT AND OUTPUT

The major inputs and outputs and major functions of the system are follows:

Input:

- The entities such as authority and users must register the account for login. All the user details have been stored the data in our database for the purpose of identifying the entities.
- The Client will upload the requirements for the report like Energy, Capacity, and Operational Hours etc.

Output:

- Only the authorized data users can able to access the data.
- Gravity Energy Storage Report will be generated at the given time.

2.5 INPUT DESIGN

- Input design is a part of overall system design. The main objective during the input design as given below.
- Input States: User can maintain a database in MySQL server or sql server for his/her business requirement.
- Input Media:

  At this stage choice has to be made about the input media. To conclude about the input media consideration has to be given to:
- In this section user can give the input for storage location and get the output from admin side.

2.6 LIMITATIONS

- Inefficiencies: The lack of an integrated system leads to inefficiencies in data processing and decision-making, as stakeholders resort to disparate tools and manual methods.

- Delays: Disjointed processes result in delays across different stages of energy management, impacting the timely execution of tasks such as client interactions, data analysis, and grid structure construction.

- Limited Transparency: The absence of a streamlined platform contributes to limited transparency in the overall energy management workflow, making it challenging for stakeholders to have a clear view of the entire process.

- Dependency on Manual Methods: Stakeholders currently heavily rely on manual methods for handling client requirements, data analysis, and maintenance tasks, introducing the potential for errors and increasing the workload.

- Lack of Awareness: The current state of disjointed processes indicates a significant gap in awareness and utilization of advanced methodologies, hindering the adoption of more efficient practices in the energy industry.

2.7 PROBLEMS IN EXISTING SYSTEM:

As of now, there is no dedicated and integrated system in place to manage the entire spectrum of energy infrastructure, from interactions to grid model construction and maintenance. The absence of a streamlined platform results in disjointed processes, hindering efficient data processing and decision-making across different stages of energy management. This lack of cohesion leads to inefficiencies, delays, and limited transparency in the overall energy management workflow. Consequently, there is a significant gap in awareness and utilization of advanced methodologies, emphasizing the need for a comprehensive energy management system like the proposed project to revolutionize and optimize current practices. The project's

exploration is timely and critical in addressing these gaps, providing a unified solution for the evolving needs of the energy industry.

2.8 PROPOSED SYSTEM

The proposed Gravity Project aims to overcome the existing disadvantages by introducing a comprehensive energy management system that addresses the entire spectrum of energy infrastructure challenges, incorporating the Advanced Algorithms like Cat Boost Regressor, MLP Regressor and Random Forest Regressor for enhanced data analysis. The system streamlines interactions, offering a user-centric environment for seamless data uploads and progress tracking. It efficiently analyzes client-uploaded energy sources, leveraging the advantages of the algorithms to make informed, data-driven decisions. Additionally, the system integrates solid mass data, eliminating delays and inefficiencies in grid model construction. This holistic approach enhances transparency and facilitates informed practices for ongoing maintenance. The central authority of the system ensures a unified approach, eliminating the need for disparate tools and manual methods. By incorporating the Algorithms, the cohesive system not only bridges gaps in awareness but also optimizes energy management practices. This not only revolutionizes the industry by providing stakeholders with a transparent, efficient, and user-friendly platform but also establishes the system as a forward-thinking solution for evolving energy landscapes. The Algorithms, with its capabilities, further enhances the effectiveness of the energy management system, making it a powerful tool for addressing current challenges and future advancements in the field.

Advantage of Proposed System:

- Streamlined Processes: The Gravity Project introduces a unified platform, streamlining processes from client interactions to grid model construction and maintenance.
- Efficient Data Processing: The system facilitates efficient data processing and decision-making, reducing delays and enhancing overall workflow.
- Reduced Manual Dependency: The project reduces dependency on manual methods, minimizing errors and decreasing the workload for stakeholders.

- Awareness and Adoption: By addressing existing gaps, the project promotes awareness and facilitates the adoption of advanced methodologies in the energy industry.
- Timely Execution: The Gravity Project ensures timely execution of tasks, from client interactions to grid model construction, optimizing overall project timelines.
- Optimized Energy Infrastructure Management: The cohesive system contributes to optimized energy infrastructure management, leveraging technological advancements for increased efficiency.
- User-Friendly Interface: The user-centric design of the system provides a friendly interface for clients, enhancing their experience in uploading data and tracking project progress.

CHAPTER-3

FEASIBILITY REPORT

3.1 Technical Feasibility

- Streamlined Processes: Integrating various functions into a single platform reduces complexity and enhances efficiency.
- Enhanced Transparency: Users can easily track and understand processes, leading to improved decision-making.
- Reduced Manual Dependencies: Automation of tasks reduces errors and frees up resources for more critical functions.
- Scalability: Designed for scalability, the system can easily integrate new features and functionalities as needed.

3.2 Operational Feasibility

- Empowerment: Comprehensive training programs and support systems empower stakeholders to effectively utilize the platform.
- Environmental Impact: By optimizing resource usage and promoting sustainable practices, the project contributes to environmental conservation.
- Collaboration: Collaboration with industry experts fosters knowledge sharing and innovation, benefiting the industry as a whole.

3.3 Economic Feasibility

- Economic benefits of aeroponics, such as increased crop yields and resource efficiency, suggest a viable economic model.
- The web application project can attract investment and support for scaling up aeroponic cultivation systems, indicating potential economic viability.
- Insights into the cost-effectiveness of aeroponics compared to traditional methods encourage adoption among farmers and agricultural businesses, highlighting economic feasibility.

CHAPTER-4

SOFTWARE REQUIREMENT SPECIFICATION

4.1 INTRODUCTION

The purpose of this document is to present a detailed description of the Web application system. It will explain the purpose and features of the system, the interfaces of the system, what the system will do, the constraints under which it must operate and how the system will react to external stimuli. This document is intended for both the stakeholders and the developers of the system and will be proposed to the Regional Historical Society for its approval.

PURPOSE

The purpose of this Software Requirement Specification (SRS) is to help the project. It is provided with some requirements which are used in the Transaction Mercator System. All parts; design, coding and testing will be prepared with helping of SRS. The purpose of this document is to detail the requirements placed on the Transaction Mercator System and serves as a contract between the customer and the developers as to what is to be expected of the stock exchange, and how the components of the system are working with each other with external systems.

This document will be checked by the group member's supervisor and it will corrected by members if supervisor orders.

DEVELOPERS RESPONSIBILITIES OVERVIEW:

The developer is responsible for:

- Developing the system, which meets the SRS and solving all the requirements of the system?
- Demonstrating the system and installing the system at client's location after the acceptance testing is successful.
- Submitting the required user manual describing the system interfaces to work on it and also the documents of the system.
- Conducting any user training that might be needed for using the system.
- Maintaining the system for a period of one year after installation.

4.1 FUNCTIONAL REQUIREMENTS:

- Following is a list of functionalities of the browsing enabled system.
- An Activity with a UI that allows you to browser settings. Provide a second Activity that allows users to access the share with permission from the administrator. Handle activity lifecycle appropriately. A precondition for any points in this part of the grade is code that compiles and runs.
- Your application should allow a user to browse the shares, buy and sell the shares with specific metadata. The assignment requires you to create a UI for browsing and a UI for integrating the two.
- The Net beans provide a number of useful layout components, views, and tools that you may want to use to create your location browser. As with the final project, you should design your application to only use the buttons on the Key board and mouse as input. Your application should use the Key board, Mouse and keywords.

4.2 NON-FUNCTIONAL REQUIREMENTS:

- The system should be supported Net beans. The member should use the System browser. Each member should have a separate system.
- The system should ask the username and password to open the application. It doesn't permit to unregistered user to access the System.
- The system should have Role based System functions access. Approval Process has to be defined.
- The system should have Modular customization components so that they can be reused across the implementation.
- These are the mainly following:
- Secure access of confidential data. 24 X 7 availability
- Better component design to get better performance at peak time
- Flexible service based architecture will be highly desirable for future extension

4.3 PERFORMANCE REQUIREMENTS

Performance is measured in terms of the output provided by the application. Requirement specification plays an important part in the analysis of a system. Only when the requirement specifications are properly given, it is possible to design a system, which will fit into required environment. It rests largely in the part of the users of the existing system to give the required specifications because they are the people who finally use the system. This is because the requirements have to be known during the initial stages so that the system can be designed according to those requirements. It is very difficult to change the system once it has been designed and on the other hand designing a system, which does not cater to the requirements of the user, is of no use.

The requirement specification for any system can be broadly stated as given below:

- The system should be able to interface with the existing system
- The system should be accurate
- The system should be better than the existing system

The existing system is completely dependent on the user to perform all the duties.

CHAPTER-5

SYSTEM DEVELOPEMENT ENVIRONMENT

5.1 INTRODUCTION TO PYTHON

About Python:

- Python was conceived in the late 1980s by Guido van Rossum at Centrum Wiskunde & Informatics in the Netherlands as a successor to ABC programming language, which was inspired by SETL, capable of exception handling and interfacing with the Amoeba operating system. Its implementation began in December 1989; Van Rossum shouldered sole responsibility for the project, as the lead developer, until 12 July 2018, when he announced his "permanent vacation" from his responsibilities as Python's Benevolent Dictator for Life, a title the Python community bestowed upon him to reflect his long-term commitment as the project's chief decision-maker. He now shares his leadership as a member of a five-person steering council. In January 2019, active Python core developers elected Brett Cannon, Nick Coughlan, Barry Warsaw, Carol Willing and Van Rossum to a five-member "Steering Council" to lead the project. Guido van Rossum has since then withdrawn his nomination for the 2020 Steering council.
- Python 2.0 was released on 16 October 2000 with many major new features, including a cycle-detecting garbage collector and support for Unicode.
- Python 3.0 was released on 3 December 2008. It was a major revision of the language that is not completely backward-compatible. Many of its major features were backported to Python 2.6.xand 2.7.x version series. Releases of Python 3 include the 2to3 utility, which automates (at least partially) the translation of Python 2 code to Python 3.
- Python 2.7's end-of-life date was initially set at 2015 then postponed to 2020 out of concern that a large body of existing code could not easily be forward-ported to Python 3. No more security patches or other improvements will be released for it. With Python 2's end-of-life, only Python 3.6.xand later are supported.
- Python 3.9.2 and 3.8.8 were expeditedas all versions of Python (including 2.7) had security issues, leading to possible remote code execution and web cache poisoning.

Key points:

- Python works on different platforms (Windows, Mac, Linux, Raspberry Pi, etc).
- Python has a simple syntax similar to the English language.
- Python has syntax that allows developers to write programs with fewer lines than some other programming languages.
- Python runs on an interpreter system, meaning that code can be executed as soon as it is written. This means that prototyping can be very quick.
- Python can be treated in a procedural way, an object-oriented way or a functional way.

Python Syntax compared to other programming languages

- Python was designed for readability, and has some similarities to the English language with influence from mathematics.
- Python uses new lines to complete a command, as opposed to other programming languages which often use semicolons or parentheses.
- Python relies on indentation, using whitespace, to define scope; such as the scope of loops, functions and classes. Other programming languages often use curly-brackets for this purpose.

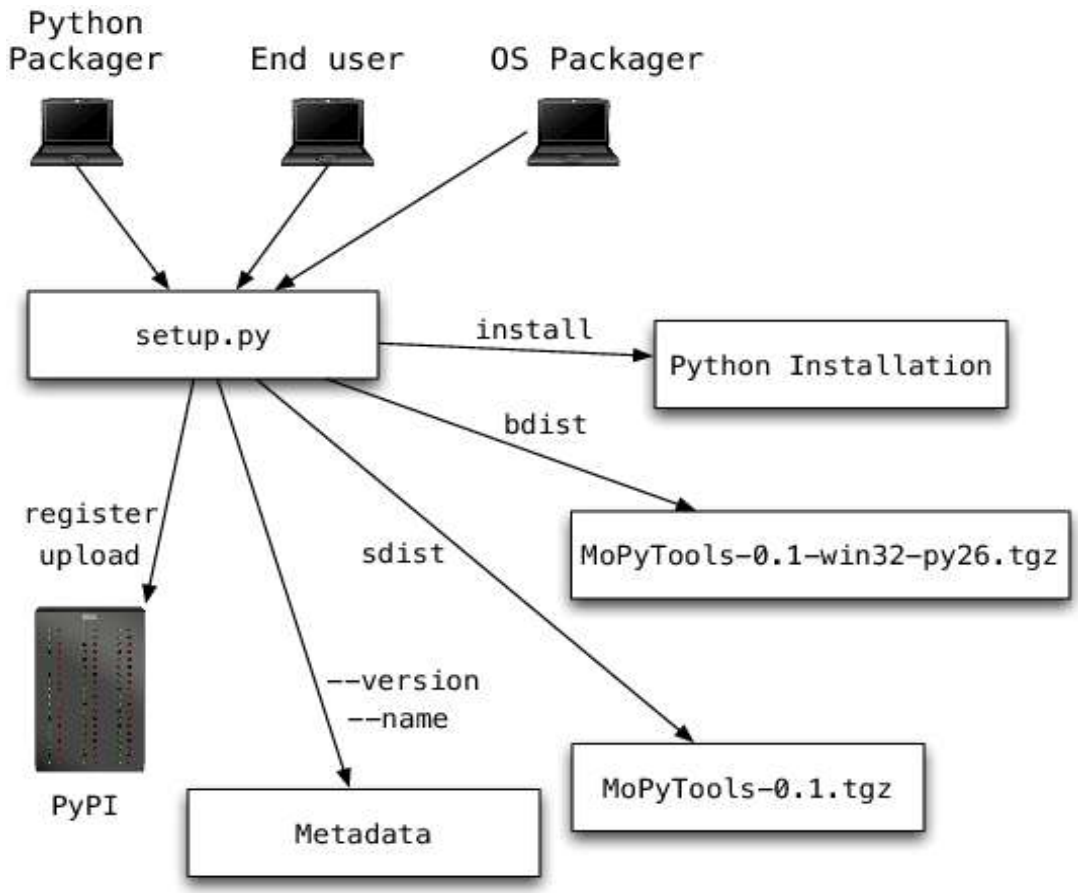
Importance of python to the Internet

Python is a general-purpose language — sometimes referred to as utilitarian — which is designed to be simple to read and write. The point that it's not a complex language is important. The designers placed less of an emphasis on conventional syntax, which makes it easier to work with,

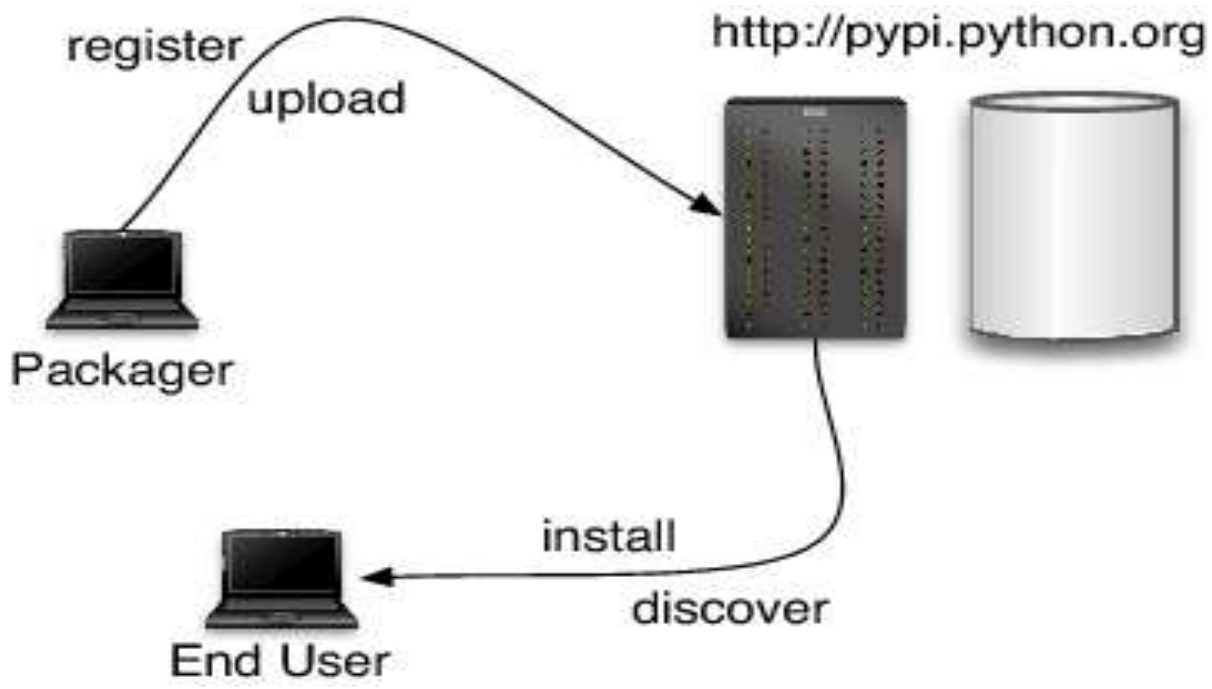
even for non-programmers or developers.Furthermore, because it's considered truly universal and used to meet various development needs, it's a language that offers a lot of options to programmers in general. If they begin working with Python for one job or career, they can easily jump to another, even if it's in an unrelated industry. The language is used for system operations, web development, server and administrative tools, deployment, scientific modelling and much

more.But, surprisingly, many developers don't pick up Python as their primary language. Because it's so easy to use and learn, they choose it as a second or third language. This may be another reason why it's so popular among developers.Plus, it just so happens that one of the biggest tech companies in the world — Google — uses the language for a number of their applications. They even have a developer portal devoted to Python, with free classes offered including exercises, lecture videos and more.In addition, the rise in the use of the Django framework for web development and a decline in popularity of PHP has also contributed to Python's success, but, ultimately, it's the perfect storm — just the right amount of developer and official support, as well as demand.

Design of python:



Python Architecture



PyPy's *Python Interpreter* is written in Python and implements the full Python language. This interpreter very closely emulates the behaviour of Python. It contains the following key components:

- a bytecode compiler responsible for producing Python code objects from the source code of a user application;
- a bytecode evaluator responsible for interpreting Python code objects;
- a standard object space, responsible for creating and manipulating the Python objects seen by the application.

The *bytecode compiler* is the pre-processing phase that produces a compact bytecode format via a chain of flexible passes (tokenizer, lexer, parser, abstract syntax tree builder, bytecode generator). The *bytecode evaluator* interprets this bytecode. It does most of its work by delegating all actual manipulations of user objects to the *object space*. The latter can be thought of as the library of built-in types. It defines the implementation of the user objects, like integers

and lists, as well as the operations between them, like addition or truth-value-testing.This division between bytecode evaluator and object space gives a lot of flexibility. One can plug in different object spaces to get different or enriched behaviours of the Python objects.

Simple:

Python was designed to be easy for the Professional programmer to learn and to use effectively. If you are an experienced C++ Programmer. Learning Python will oriented features of C++. Most of the confusing concepts from C++ are either left out of Java or implemented in a cleaner, more approachable manner. In Java there are a small number of clearly defined ways to accomplish a given task.

Object oriented

Python was not designed to be source-code compatible with any other language. This allowed the Python team the freedom to design with a blank state. One outcome of this was a clean, usable, pragmatic approach to objects. The object model in Python is simple and easy to extend, while simple types, such as integers, are kept as high-performance non-objects.

Robust

The multi-platform environment of the web places extraordinary demands on a program, because the program must execute reliably in a variety of systems. The ability to create robust programs.Was given a high priority in the design of Python. Python is strictly typed language; it checks your code at compile time and runtime.

Python virtually eliminates the problems of memory management and deal location, which is completely automatic. In a well-written Python program, all run-time errors can and should be managed by your program.

USER AUTHORIZATION:

One way to perform session tracking is to leverage the information that comes with User authorization. When a web server restricts access to some of its resources to only those clients that log in using a recognized username and password. After the client logs in, the username is available to a servlet through getRemoteUser ().

When use the username to track the session. Once a user has logged in, the browser remembers her user name and resends the name and password as the user views new pages on the site. A servlet can identify the user through her username and they're by Track her session.

The biggest advantage of using user authorization to perform session tracking is that it's easy to implement. Simply tell the protect a set of pages, and use getRemoteUser() to identify each client. Another advantage is that the technique works even when the user accesses your site form or exits her browser before coming back.

The biggest disadvantage of user authorization is that it requires each user to register for an account and then log in each time the starts visiting your site. Most users will tolerate registering and logging in as a necessary evil when they are accessing sensitive information, but it's all overkill for simple session tracking. Another problem with user authorization is that a user cannot simultaneously maintain more than one session at the same site.

Hidden Form Fields:

One way to support anonymous session tracking is to use hidden from the fields. As the name implies, these are fields added to an HTML, from that are not displayed in the client's browser, they are sent back to the server when the form that contains them is submitted.

In a sense, hidden form fields define constant variables for a form. For a servlet receiving a submitted form, there is no difference between a hidden field and a visible filed.

As more information is associated with a client's session. It can become burdensome to pass it all using hidden form fields. In these situations, it's possible to pass on just a unique session ID that identifies as a particular client session.

That session ID can be associated with complete information about its session that is stored on the server.

The advantage of hidden form fields is their ubiquity and support for anonymity. Hidden fields are supported in all the popular browsers, they demand on special server requirements, and they can be used with clients that haven't registered or logged in.

The major disadvantage with this technique, however, is that works only for a sequence of dynamically generated forms, the technique breaks down immediately with static documents, emailed documents bookmarked documents and browser shutdowns.

URL Rewriting:

URL rewriting is another way to support anonymous session tracking, With URL rewriting every local URL the user might click on is dynamically modified. Or rewritten, to include extra, information. The extra information can be in the form of extra path information, added parameters, or some custom, server-specific.URL change. Due to the limited space available in rewriting a URL, the extra information is usually limited to a unique session.

Each rewriting technique has its own advantage and disadvantage

Using extra path information works on all servers, and it works as a target for forms that use both the Get and Post methods. It does not work well if the servlet has to use the extra path information as true path information

The advantages and disadvantages of URL. Rewriting closely match those of hidden form fields, The major difference is that URL rewriting works for all dynamically created documents, such as the Help servlet, not just forms. With the right server support, custom URL rewriting can even work for static documents.

Persistent Cookies:

A fourth technique to perform session tracking involves persistent cookies. A cookie is a bit of information. Sent by a web server to a browser that can later be read back form that browser.

When a browser receives a cookie, it saves the cookie and there after sending the cookie back to the server each time it accesses a page on that server, subject to certain rules. Because a cookie's value can uniquely identify a client, cookies are often used for session tracking.

Persistent cookies offer an elegant, efficient, easy way to implement session tracking. Cookies provide as automatic an introduction for each request, as we could hope for. For each request, a cookie can automatically provide a client's session ID or perhaps a list of clients' performance. The ability to customize cookies gives them extra power and versatility.

The biggest problem with cookies is that browsers don't always accept cookies sometimes this is because the browser doesn't support cookies. More often it's because the browser doesn't support cookies. More often it's because the user has specifically configured the browser to refuse cookies.

Languages influenced by Python

Python's design and philosophy have influenced many other programming languages:

- Boo uses indentation, a similar syntax, and a similar object model.
- Cobra uses indentation and a similar syntax, and its *Acknowledgements* document lists Python first among languages that influenced it.
- Coffee Script, a programming language that cross-compiles to JavaScript, has Python-inspired syntax.
- ECMAScript/JavaScript borrowed iterators and generators from Python.
- GD Script, a scripting language very similar to Python, built-in to the Godot game engine.
- Go is designed for the "speed of working in a dynamic language like Python"and shares the same syntax for slicing arrays.
- Groovy was motivated by the desire to bring the Python design philosophy to Java.
- Julia was designed to be "as usable for general programming as Python".
- Nim uses indentation and similar syntax.

- Ruby's creator, Yukihiro Matsumoto, has said: "I wanted a scripting language that was more powerful than Perl, and more object-oriented than Python. That's why I decided to design my own language.
- Swift, a programming language developed by Apple, has some Python-inspired syntax.

Python's development practices have also been emulated by other languages. For example, the practice of requiring a document describing the rationale for, and issues surrounding, a change to the language (in Python, a PEP) is also used in Tcl, Erlang, and Swift.

Django:

Django sometimes stylized as **django** is a Python-based free and open-source web framework that follows the model-template-views (MTV) architectural pattern. It is maintained by the Django Software Foundation (DSF), an American independent organization established as non-profit.Django's primary goal is to ease the creation of complex, database-driven websites. The framework emphasizes reusability of components, less code, low coupling, rapid development, and the principle of don't repeat yourself.Python is used throughout, even for settings, files, and data models. Django also provides an optional administrative create, read, update and delete interface that is generated dynamically through introspection and configured via admin models. Despite having its own nomenclature, such as naming the callable objects generating the HTTP responses "views",[9] the core Django framework can be seen as an MVC architecture.[10] It consists of an object-relational mapper (ORM) that mediates between data models (defined as Python classes) and a relational database ("**M**odel"), a system for processing HTTP requests with a web templating system ("**V**iew"), and a regular-expression-based URL dispatcher ("**C**ontroller")

Flask:

Flask is a micro web framework written in Python. It is classified as a microframework because it does not require particular tools or libraries.It has no database abstraction layer, form validation, or any other components where pre-existing third-party libraries provide common functions. However, Flask supports extensions that can add application features as if they were implemented in Flask itself. Extensions exist for object-relational mappers, form validation, upload handling, various open authentication technologies and several common framework related tools.

The microframework Flask is based on the Pocoo projects, Werkzeug and Jinja2.

Werkzeug

Werkzeug is a utility library for the Python programming language, in other words a toolkit for Web Server Gateway Interface (WSGI) applications, and is licensed under a BSD License. Werkzeug can realize software objects for request, response, and utility functions. It can be used to build a custom software framework on top of it and supports Python 2.7 and 3.5 and later.

Jinja

Jinja, also by Ronacher, is a template engine for the Python programming language and is licensed under a BSD License. Similar to the Django web framework, it handles templates in a sandbox.

5.3 HTML, JAVASCRIPT

Hypertext Markup Language(HTML), the languages of the World Wide Web (WWW), allows users to produce web pages that included text, graphics and pointer to other web pages (Hyperlinks).

HTML is not a programming language, but it is an application of ISO Standard 8879, SGML (Standard Generalized Markup Language), but Specialized to hypertext and adapted to the Web. The idea behind Hypertext one point to another point. We can navigate through the information based on out interest and preference. A markup language is simply a series of items

enclosed within the elements should be displayed.Hyperlinks are underlined or emphasized works that load to other documents or some portions of the same document.

HTML can be used to display any type of document on the host computer, which can be geographically at a different location. It is a versatile language and can be used on any platform or desktop

HTML provides tags (special codes) to make the document look attractive.

HTML provides are not case-sensitive. Using graphics, fonts, different sizes, color, etc., can enhance the presentation of the document. Anything that is not a tag is part of the document itself.

Basic Html Tags:

<!-- -->	Specific Comments.
<A>……</A>	Creates Hypertext links.
<B>……</B>	Creates hypertext links.
<Big>……..</Big>	Formats text in large-font
<Body>……. </Body>	contains all tags and text in the Html-document
<Center>……</Center>	Creates Text
<DD>……..</DD>	Definition of a term.
<TABLE>…… </TABLE>	creates a table
<Td>……..</Td>	indicates table data in a table.
<Tr>……..</Tr>	designates a table row
<Th>……. </Th>	creates a heading in a table.

ADVANTAGE

- A HTML document is small and hence easy to send over the net. It is small because it does not include formatted information.
- HTML is platform independent
- HTML tags are not case-sensitive.

5.4 JAVA SCRIPT

JavaScript is a compact, object-based scripting language for developing client and server internet applications. Netscape Navigator 2.0 interprets JavaScript statements embedded directly in an HTML page. And Livewire enables you to create server-based applications similar to common gateway interface (CGI) programs.

In a client application for Navigator, JavaScript statements embedded in an HTML Page can recognize and respond to user events such as mouse clicks form input, and page navigation.

For example, you can write a JavaScript function to verify that users enter valid information into a form requesting a telephone number or zip code. Without any network transmission, an Html page with embedded Java Script can interpret the entered text and alert the user with a message dialog if the input is invalid or you can use JavaScript to perform an action (such as play an audio file, execute an applet, or communicate with a plug-in) in response to the user opening or exiting a page.

CHAPTER-6

SYSTEM DESIGN

6.1 INTRODUCTION

The project "Celestial Path Exploration through a Multifaceted Simulation System for Trajectory Analysis of Celestial Bodies" is a groundbreaking initiative designed to revolutionize energy data processing and grid model construction. By integrating a user-friendly interface, the system caters to diverse user roles, ensuring easy registration and submission of data sources. Transparency is a key focus, with the system facilitating effective communication and evaluation of submissions. Team members contribute to the system by uploading energy data sources, which are then used to establish a robust foundation for grid model construction. The integration of power source and solid mass data provides crucial insights for precise modeling, while the incorporation of Solid Gravity Energy data enhances system maintenance practices. The project emphasizes sustainability by enabling users to identify consumption patterns and optimize resource usage. Its user-centric design enhances overall energy infrastructure management, fostering effectiveness and user satisfaction. The system's adaptability to emerging technologies ensures its relevance in the face of industry advancements, providing a future-proof solution for evolving energy landscapes. Overall, the project not only revolutionizes energy data processing and grid model construction but also promotes environmental responsibility and technological progress in energy infrastructure management.

MODULES:

1) Client
2) Power Source
3) Grid Structure
4) System Maintenance
5) Admin

MODULE 1: CLIENT

This module streamlines the client registration process, providing access to a dedicated client page where users can log in or create a new account if they are new to the platform. Once logged in, clients are empowered to upload their requirement data source through a user-friendly interface. A proficient administrator with full administrative privileges meticulously assesses the client's information and the requirement data source, taking prompt actions such as approving or rejecting the client as necessary. Clear and timely communication is maintained with clients via email to ensure they are well-informed about their project status. Additionally, clients can track the progress of their requirements and explore a variety of final processed reports after all the processing stages are completed. Moreover, clients are required to make payments for their process and final report, contributing to an ever-evolving, user-centric environment that continually enhances the platform's effectiveness and user satisfaction. The module incorporates robust security measures to safeguard sensitive client data, ensuring confidentiality throughout the registration and data upload processes. This commitment to data security underscores our dedication to providing a trustworthy and secure environment for our valued clients.

MODULE 2: POWER SOURCE

The Power Source team gains significant advantages through a streamlined and efficient access system to the Power Source portal, facilitated by a registration and approval process administered by the system administrator. Upon approval, team members can securely log in

using their designated email and password, with the password sent to their registered email by the administrator. The functionality of this module relies on the requirement data sources uploaded by the client. The Power Source team is empowered to upload energy data sources based on specific requirements, facilitating the determination of average energy generation. The calculated results can then be seamlessly transmitted to the Grid Structure team, initiating subsequent processes with a well-informed and data-driven foundation. This cohesive workflow ensures a smooth and organized approach to managing and utilizing power source data within the broader operational framework. The system incorporates real-time collaboration features, enabling seamless communication among Power Source team members and fostering collaborative decision-making for enhanced efficiency in energy data analysis. Additionally, regular updates and alerts keep team members informed of critical milestones, ensuring proactive engagement and timely responses to evolving energy generation dynamics.

MODULE 3: GRID STRUCTURE

The Grid Structure team gains significant advantages through a streamlined and efficient access system to the Grid Structure portal, facilitated by a registration and approval process administered by the system administrator. Upon approval, Grid Structure team members can securely log in using their designated email and password, with the password sent to their registered email by the administrator. In response to the power source report submitted by the Power Source team, the Grid Structure team initiates the upload of solid mass data sources, aligning them with client specifications. The solid mass data is intricately mapped with the power source report, yielding critical insights for constructing the grid model. Subsequently, the comprehensive Grid Structure report is dispatched to the System Maintenance team. This collaborative workflow ensures the seamless integration of power source and solid mass data, contributing to both precise grid model construction and informed system maintenance practices. The Grid Structure portal features interactive visualization tools, empowering team members to explore models and simulate grid scenarios, enhancing their understanding and decision-making capabilities. Regular training sessions are conducted to keep the team abreast of the latest advancements in grid modeling techniques, fostering continuous improvement and expertise within the Grid Structure team.

MODULE 4: SYSTEM MAINTENANCE

The System Maintenance team gains significant advantages through a streamlined and efficient access system to the System Maintenance portal, facilitated by a registration and approval process administered by the system administrator. Upon approval, System Maintenance team members can securely log in using their designated email and password, with the password sent to their registered email by the administrator. In response to the Grid Structure report submitted by the Grid Structure team, the System Maintenance team initiates the upload of Solid Gravity Energy data sources, aligning them with client specifications. The Solid Gravity Energy data is intricately mapped with the grid structure report, yielding critical insights for maintenance of the grid model. Subsequently, the comprehensive System Maintenance report is dispatched to the Client. This collaborative workflow ensures the seamless integration of Grid structure and Solid Gravity Energy, contributing to both precise grid model construction and informed system maintenance practices. The final report sent to the administration and the client. The System Maintenance portal incorporates automated alerts and predictive analytics, enabling the team to proactively address potential issues and schedule preventive maintenance, ensuring the reliability and longevity of the grid system.

MODULE 5: ADMIN

The administrator holds exclusive access to the admin panel, leveraging a default username and password for authentication. This panel provides the administrator with the authority to thoroughly evaluate and make informed decisions on the requirement data sources submitted by clients, allowing them to either approve or reject the submissions. Upon client approval, an automated email notification promptly informs clients, fostering transparency and communication throughout the resolution process. The administrator has the capability to review the conclusive outcomes from the power source, grid structure, and system maintenance. Subsequently, billing is executed by the administrator based on the assessed results. Positioned as the ultimate authority, the admin serves as the central controller orchestrating the entire process. The admin panel offers advanced analytics tools, enabling the administrator to extract valuable insights from aggregated data, identify trends, and optimize the overall operational

workflow. Additionally, the administrator plays a pivotal role in refining platform policies and procedures, ensuring continuous alignment with industry standards and client expectations for a dynamic and responsive user experience.

Database Screen Short:

localhost » gravity » client_registration

Browse | Structure | SQL | Search | Insert | Export | Import | Operations | Triggers

#	Name	Type	Collation	Attributes	Null	Default	Extra	Action
1	id	bigint(20)			No	None	AUTO_INCREMENT	Change Drop Primary Unique Index Spatial More
2	client_id	varchar(20)	latin1_swedish_ci		Yes	NULL		Change Drop Primary Unique Index Spatial More
3	companyname	varchar(200)	latin1_swedish_ci		No	None		Change Drop Primary Unique Index Spatial More
4	founder	varchar(200)	latin1_swedish_ci		No	None		Change Drop Primary Unique Index Spatial More
5	address	varchar(200)	latin1_swedish_ci		No	None		Change Drop Primary Unique Index Spatial More
6	email	varchar(254)	latin1_swedish_ci		Yes	NULL		Change Drop Primary Unique Index Spatial More
7	approve	tinyint(1)			Yes	NULL		Change Drop Primary Unique Index Spatial More
8	reject	tinyint(1)			Yes	NULL		Change Drop Primary Unique Index Spatial More
9	password	bigint(20)		Unsigned	Yes	NULL		Change Drop Primary Unique Index Spatial More
10	login	tinyint(1)			Yes	NULL		Change Drop Primary Unique Index Spatial More
11	logout	tinyint(1)			Yes	NULL		Change Drop Primary Unique Index Spatial More
12	upload	tinyint(1)			Yes	NULL		Change Drop Primary Unique Index Spatial More
13	status	varchar(10)	latin1_swedish_ci		No	None		Change Drop Primary Unique Index Spatial More
14	order_id	varchar(20)	latin1_swedish_ci		Yes	NULL		Change Drop Primary Unique Index Spatial More
15	final	tinyint(1)			Yes	NULL		Change Drop Primary Unique Index Spatial More
16	final1	tinyint(1)			Yes	NULL		Change Drop Primary Unique Index Spatial More
17	end	tinyint(1)			Yes	NULL		Change Drop Primary Unique Index Spatial More
18	amount	varchar(10)	latin1_swedish_ci		Yes	NULL		Change Drop Primary Unique Index Spatial More
19	pay	tinyint(1)			Yes	NULL		Change Drop Primary Unique Index Spatial More
20	process	tinyint(1)			Yes	NULL		Change Drop Primary Unique Index Spatial More

localhost » gravity » client_equipment

Browse | Structure | SQL | Search | Insert | Export | Import | Operations | Triggers

#	Name	Type	Collation	Attributes	Null	Default	Extra	Action
1	id	bigint(20)			No	None	AUTO_INCREMENT	Change Drop Primary More
2	client_id	varchar(20)	latin1_swedish_ci		Yes	NULL		Change Drop Primary More
3	month	varchar(20)	latin1_swedish_ci		Yes	NULL		Change Drop Primary More
4	energy	varchar(20)	latin1_swedish_ci		Yes	NULL		Change Drop Primary More
5	turbine_capacity	double			Yes	NULL		Change Drop Primary More
6	energy_generated	double			Yes	NULL		Change Drop Primary More
7	capacity_factor	double			Yes	NULL		Change Drop Primary More
8	operation_hours	int(11)			Yes	NULL		Change Drop Primary More
9	psdone1	tinyint(1)			Yes	NULL		Change Drop Primary More
10	gravitational_potential_energy	double			Yes	NULL		Change Drop Primary More
11	energy_turbine_capacity_needed	double			Yes	NULL		Change Drop Primary More
12	hours_needed_for_energy_generation	double			Yes	NULL		Change Drop Primary More
13	turbine_area	double			Yes	NULL		Change Drop Primary More
14	gsdone2	tinyint(1)			Yes	NULL		Change Drop Primary More
15	energy_storage_capacity_mwh	double			Yes	NULL		Change Drop Primary More
16	mass_metric_tons	double			Yes	NULL		Change Drop Primary More
17	height_of_elevation_meters	double			Yes	NULL		Change Drop Primary More
18	density_kg	double			Yes	NULL		Change Drop Primary More
19	volume	double			Yes	NULL		Change Drop Primary More
20	dimensions_m	double			Yes	NULL		Change Drop Primary More
21	efficiency	double			Yes	NULL		Change Drop Primary More

localhost » gravity » client_requirement

Browse | Structure | SQL | Search | Insert | Export | Import | Operations | Triggers

22	Energy_Density_Wh	double			Yes	NULL
23	Power_Density_W	double			Yes	NULL
24	Energy_Rating_Wh	double			Yes	NULL
25	Power_Rating_W	double			Yes	NULL
26	Discharge_Time_h	double			Yes	NULL
27	material_type	varchar(40)	latin1_swedish_ci		Yes	NULL
28	structure	varchar(40)	latin1_swedish_ci		Yes	NULL
29	mechanism	varchar(40)	latin1_swedish_ci		Yes	NULL
30	control	varchar(40)	latin1_swedish_ci		Yes	NULL
31	roundtrip_efficiency	double			Yes	NULL
32	life_time_years	bigint(20)		UNSIGNED	Yes	NULL
33	regular_inspection	varchar(400)	latin1_swedish_ci		Yes	NULL
34	lubrication	varchar(400)	latin1_swedish_ci		Yes	NULL
35	cleaning_maintenance	varchar(400)	latin1_swedish_ci		Yes	NULL
36	control_system_monitoring	varchar(400)	latin1_swedish_ci		Yes	NULL
37	power_conversion_maintenance	varchar(400)	latin1_swedish_ci		Yes	NULL
38	safety_systems	varchar(400)	latin1_swedish_ci		Yes	NULL
39	environmental_considerations	varchar(400)	latin1_swedish_ci		Yes	NULL
40	material_integrity	varchar(400)	latin1_swedish_ci		Yes	NULL
41	data_monitoring_analysis	varchar(400)	latin1_swedish_ci		Yes	NULL
42	training_personnel	varchar(400)	latin1_swedish_ci		Yes	NULL
43	emergency_response	varchar(400)	latin1_swedish_ci		Yes	NULL

Change Drop Primary More

localhost » gravity » client_requirement

Browse | Structure | SQL | Search | Insert | Export | Import | Operations | Triggers

31	roundtrip_efficiency	double			Yes	NULL
32	life_time_years	bigint(20)		UNSIGNED	Yes	NULL
33	regular_inspection	varchar(400)	latin1_swedish_ci		Yes	NULL
34	lubrication	varchar(400)	latin1_swedish_ci		Yes	NULL
35	cleaning_maintenance	varchar(400)	latin1_swedish_ci		Yes	NULL
36	control_system_monitoring	varchar(400)	latin1_swedish_ci		Yes	NULL
37	power_conversion_maintenance	varchar(400)	latin1_swedish_ci		Yes	NULL
38	safety_systems	varchar(400)	latin1_swedish_ci		Yes	NULL
39	environmental_considerations	varchar(400)	latin1_swedish_ci		Yes	NULL
40	material_integrity	varchar(400)	latin1_swedish_ci		Yes	NULL
41	data_monitoring_analysis	varchar(400)	latin1_swedish_ci		Yes	NULL
42	training_personnel	varchar(400)	latin1_swedish_ci		Yes	NULL
43	emergency_response	varchar(400)	latin1_swedish_ci		Yes	NULL
44	upgrades_improvements	varchar(400)	latin1_swedish_ci		Yes	NULL
45	smdone3	tinyint(1)			Yes	NULL
46	final_report	varchar(100)	latin1_swedish_ci		Yes	NULL
47	report_view	tinyint(1)			Yes	NULL
48	finalreportapprove	tinyint(1)			Yes	NULL
49	finalreportreject	tinyint(1)			Yes	NULL
50	payamt	tinyint(1)			Yes	NULL

Change Drop Primary More

Check All With selected: Browse Change Drop Primary Unique Index

localhost » gravity » client_payment

Browse | Structure | SQL | Search | Insert | Export | Import | Operations | Triggers

#	Name	Type	Collation	Attributes	Null	Default	Extra	Action
1	id	bigint(20)			No	None	AUTO_INCREMENT	Change Drop Primary Unique Index Spatial More
2	client_id	varchar(20)	latin1_swedish_ci		Yes	NULL		Change Drop Primary Unique Index Spatial More
3	order_id	varchar(20)	latin1_swedish_ci		Yes	NULL		Change Drop Primary Unique Index Spatial More
4	amount	varchar(10)	latin1_swedish_ci		Yes	NULL		Change Drop Primary Unique Index Spatial More
5	cardholder	varchar(40)	latin1_swedish_ci		Yes	NULL		Change Drop Primary Unique Index Spatial More
6	cardnumber	varchar(40)	latin1_swedish_ci		Yes	NULL		Change Drop Primary Unique Index Spatial More
7	expirydate	varchar(10)	latin1_swedish_ci		Yes	NULL		Change Drop Primary Unique Index Spatial More
8	cvv	int(11)			Yes	NULL		Change Drop Primary Unique Index Spatial More

Check All With selected: Browse Change Drop Primary Unique Index

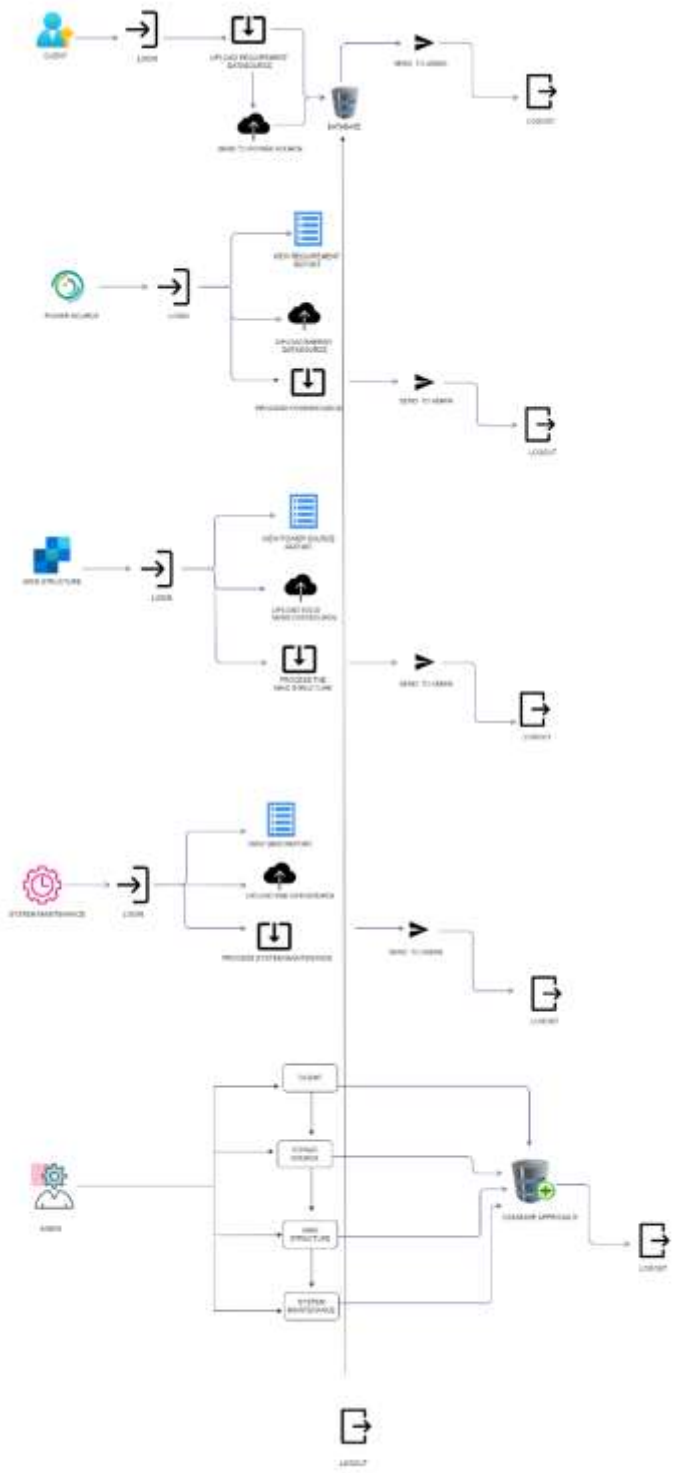
localhost » gravity » powersource_emp_details

Browse | Structure | SQL | Search | Insert | Export | Import | Operations | Triggers

#	Name	Type	Collation	Attributes	Null	Default	Extra	Action
1	id	bigint(20)			No	None	AUTO_INCREMENT	Change Drop Primary Unique Index Spatial More
2	empid	varchar(20)	latin1_swedish_ci		Yes	NULL		Change Drop Primary Unique Index Spatial More
3	empname	varchar(200)	latin1_swedish_ci		No	None		Change Drop Primary Unique Index Spatial More
4	empaddress	varchar(200)	latin1_swedish_ci		No	None		Change Drop Primary Unique Index Spatial More
5	empemail	varchar(254)	latin1_swedish_ci		No	None		Change Drop Primary Unique Index Spatial More
6	emppassword	bigint(20)		UNSIGNED	Yes	NULL		Change Drop Primary Unique Index Spatial More
7	department	varchar(200)	latin1_swedish_ci		Yes	NULL		Change Drop Primary Unique Index Spatial More
8	grant	tinyint(1)			Yes	NULL		Change Drop Primary Unique Index Spatial More
9	revoke	tinyint(1)			Yes	NULL		Change Drop Primary Unique Index Spatial More
10	accept	tinyint(1)			Yes	NULL		Change Drop Primary Unique Index Spatial More
11	decline	tinyint(1)			Yes	NULL		Change Drop Primary Unique Index Spatial More
12	admit	tinyint(1)			Yes	NULL		Change Drop Primary Unique Index Spatial More
13	deny	tinyint(1)			Yes	NULL		Change Drop Primary Unique Index Spatial More

Check All With selected: Browse Change Drop Primary Unique Index

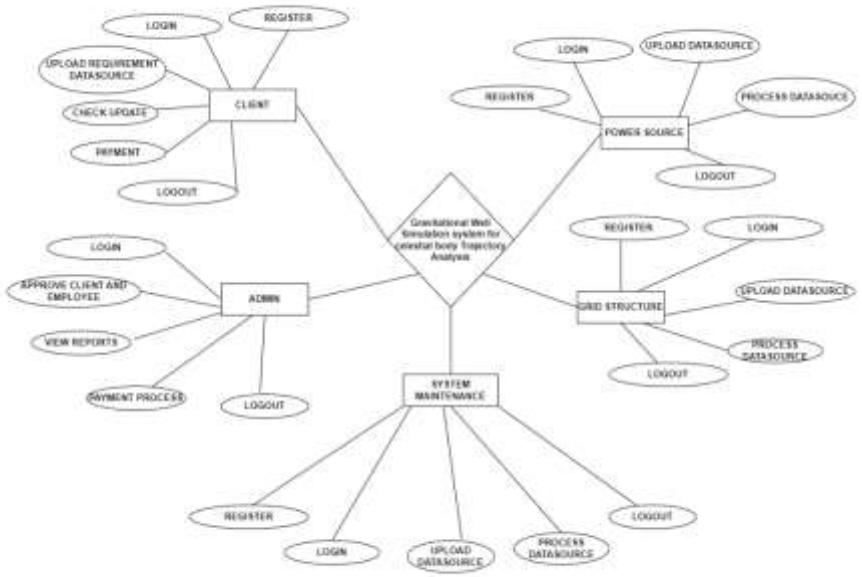
6.3 System Architecture:



6.4 E – R DIAGRAMS

- The relation upon the system is structured through a conceptual ER-Diagram, which not only specifics the existing entities, but also the standard relations through which the system exists and the cardinalities that are necessary for the system state to continue.
- The Entity Relationship Diagram (ERD) depicts the relationship between the data objects. The ERD is the notation that is used to conduct, the date modeling activity the attributes of each data object noted, is the ERD can be described resign a data object description.
- The set of primary components that are identified by the ERD are
    - Data object
    - Relationships
    - Attributes
    - Various types of indicators.

The primary purpose of the ERD is to represent data objects and their relationships.



6.5 FLOW DIAGRAMS

A data flow diagram is a graphical tool used to describe and analyze the movement of data through a system. These are the central tool and the basis from which the other components are developed. The transformation of data from input to output through processing, may be described logically and independently of physical components associated with the system. These are known as the logical data flow diagrams. The physical data flow diagrams show the actual implements and movement of data between people, departments and workstations. A full description of a system actually consists of a set of data flow diagrams. Using two familiar notations Yourdon, Gane and Sarson notation develops the data flow diagrams. Each component in a DFD is labeled with a descriptive name. The process is further identified with a number that will be used for identification purpose. The development of DFD'S is done on several levels. Each process in lower level diagrams can be broken down into a more detailed DFD in the next level. The lop-level diagram is often called context diagram. It consist a single process bit, which plays a vital role in studying the current system. The process in the context level diagram is exploded into another process at the first level DFD.

The idea behind the explosion of a process into more process is that understanding at one level of detail is exploded into greater detail at the next level. This is done until further explosion is necessary and an adequate amount of detail is described for analysts to understand the process.

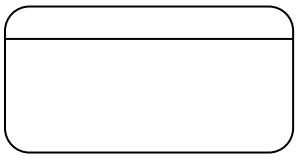
Larry Constantine first developed the DFD as a way of expressing system requirements in a graphical form, this lead to the modular design.

A DFD is also known as a "bubble Chart" has the purpose of clarifying system requirements and identifying major transformations that will become programmed in system design. So it is the starting point of the design to the lowest level of detail. A DFD consists of a series of bubbles joined by data flows in the system.

6.6 DFD SYMBOLS

In the DFD, there are four symbols

1. A square defines a source (originating) or destination of system data
2. An arrow identifies data flow. It is the pipeline through which the information flows
3. A circle or a bubble represents a process that transforms the incoming data flow into outgoing data flows.
4. An open rectangle is a data store, data at rest or a temporary repository of data



A process that transforms the data flow

Source or Destination of data

Data flow

Data Store

CONSTRUCTING A DFD:

Several rules of thumb are used in drawing DFD'S:

1. Process should be named and numbered for an easy reference. Each name should be representative of the process.
2. The direction of flow is from top to bottom and from left to right. Data traditionally flow from source to the destination although they may flow back to the source. One way to indicate this is to draw the long flow line back to a source. An alternative way is to repeat the source symbol as a destination. Since it is used more than once in the DFD it is marked with a short diagonal.

3. When a process is exploded into lower level details, they are numbered.
4. The names of data stores and destinations are written in capital letters. Process and dataflow names have the first letter of each work capitalized

A DFD typically shows the minimum contents of data store. Each data store should contain all the data elements that flow in and out.

Questionnaires should contain all the data elements that flow in and out. Missing interfaces redundancies and like is then accounted for often through interviews.

SAILENT FEATURES OF DFD'S

1. The DFD shows the flow of data, not of control loops and decision are controlled considerations do not appear on a DFD.
2. The DFD does not indicate the time factor involved in any process, whether the dataflow take place daily, weekly, monthly or yearly.
3. The sequence of events is not brought out on the DFD.

TYPES OF DATA FLOW DIAGRAMS

1. Current Physical
2. Current Logical
3. New Logical
4. New Physical

CURRENT PHYSICAL

In Current Physical DFD process label includes the name of the people or their positions or the names of computer systems that might provide some of the overall system-processing label includes an identification of the technology used to process the data. Similarly, data flows and data stores are often labelled with the names of the actual physical media on which data are stored such as file folders, computer files, business forms or computer tapes.

CURRENT LOGICAL:

The physical aspects in the system are removed as much as possible so that the current system is reduced to its essence to the data and the processes that transform them regardless of actual physical form.

NEW LOGICAL:

This is exactly like a current logical model if the user were completely happy with the user were completely happy with the functionality of the current system, but had problems with how it was implemented typically through the new logical model will differ from the current logical model while having additional functions, absolute function removal and inefficient flows recognized.

NEW PHYSICAL:

The new physical represents only the physical implementation of the new system.

RULES GOVERNING THE DFD'S

PROCESS

1) No process can have only outputs.
2) No process can have only inputs. If an object has only inputs than it must be a sink.
3) A process has a verb phrase label.

DATA STORE

1) Data cannot move directly from one data store to another data store, a process must move data.
2) Data cannot move directly from an outside source to a data store, a process, which receives, must move data from the source and place the data into the data store
3) A data store has a noun phrase label.

SOURCE OR SINK

The origin and /or destination of data

1) Data cannot move direly from a source to sink it must be moved from a process
2) A source and /or sink have a noun phrase land

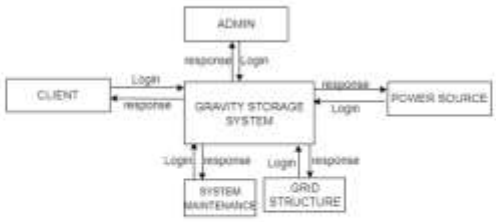
DATA FLOW

1) A Data Flow has only one direction of flow between symbols. It may flow in both directions between a process and a data store to show a read before an update. The latter is usually indicated, however by two separate arrows since these happen at different type.
2) A join in DFD means that exactly the same data comes from any of two or more different processes data store or sink to a common location.
3) A data flow cannot go directly back to the same process it leads. There must be at least one other process that handles the data flow produce some other data flow returns the original data in the beginning process.
4) A Data flow to a data store means update (delete or change).
5) A data Flow from a data store means retrieve or use.

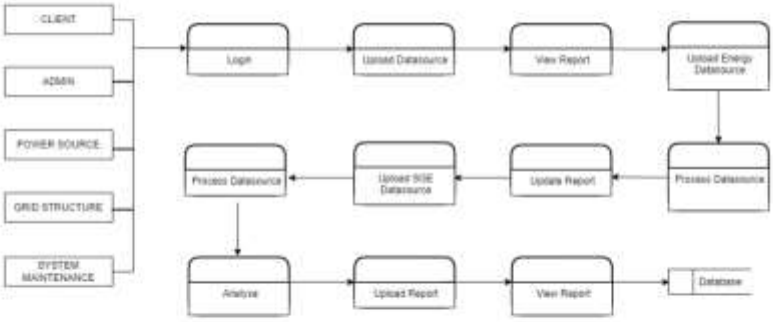
A data flow has a noun phrase label more than one data flow noun phrase can appear on a single arrow as long as all of the flows on the same arrow move together as one package.

DATA FLOW DIAGRAM

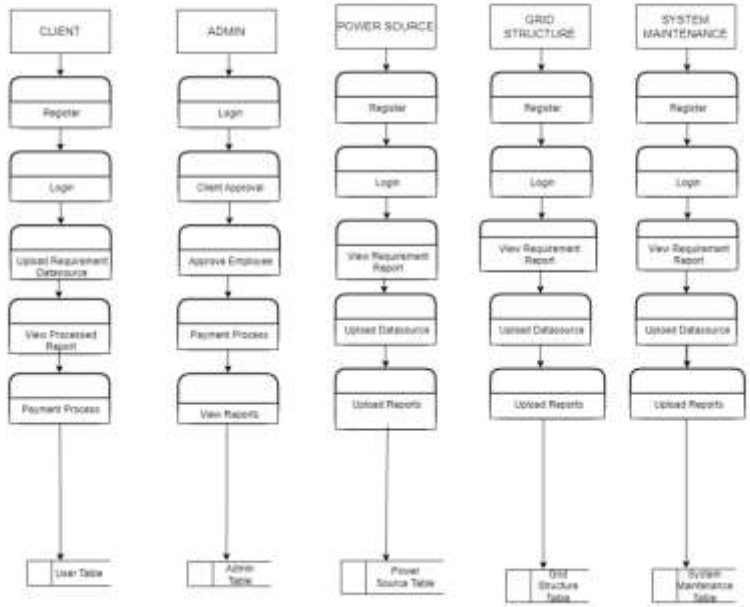
Level 0



Level 1

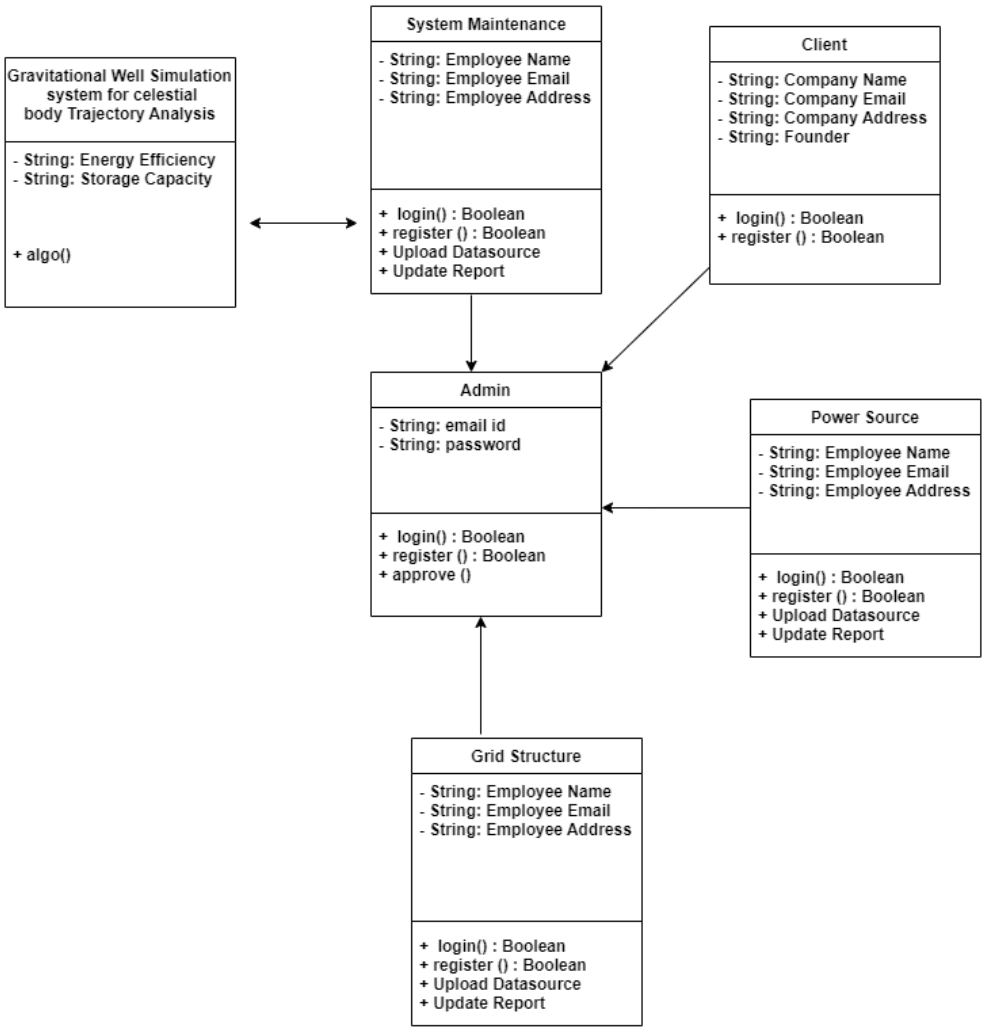


Level 2



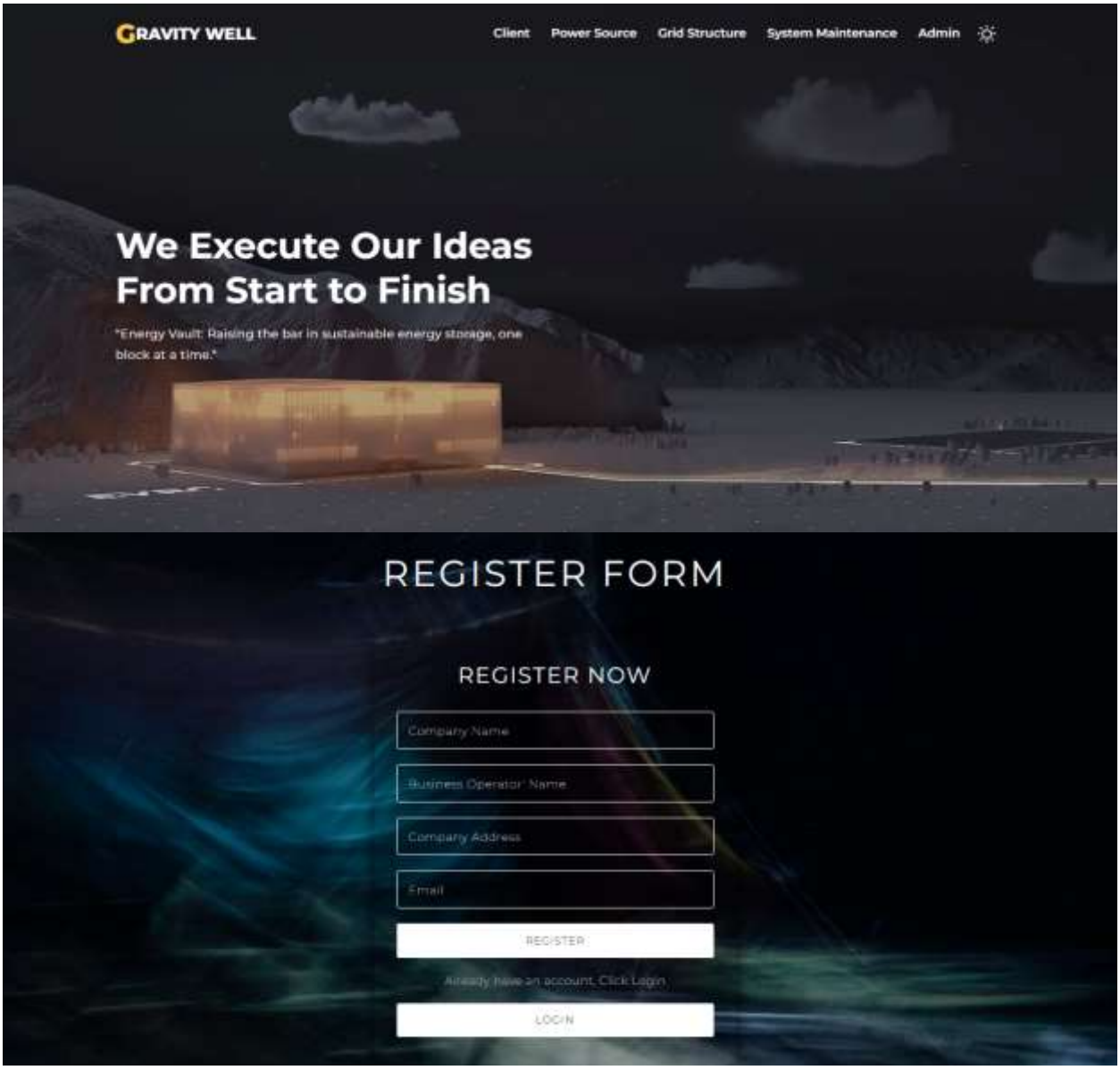
6.8 USE CASE DIAGRAM

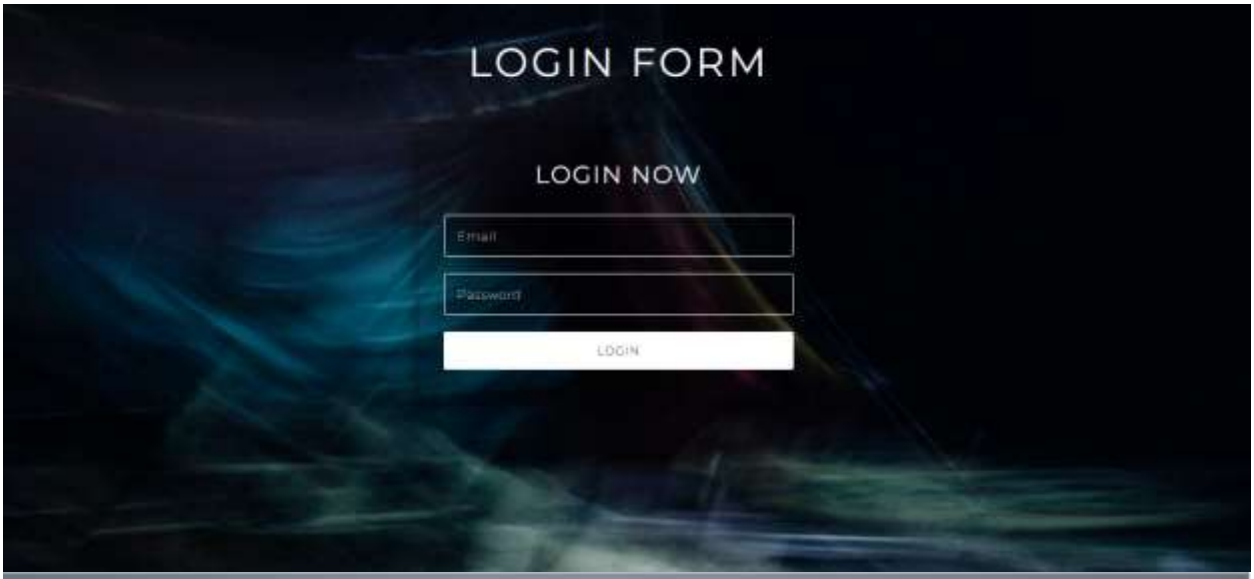
6.10 CLASS DIAGRAM

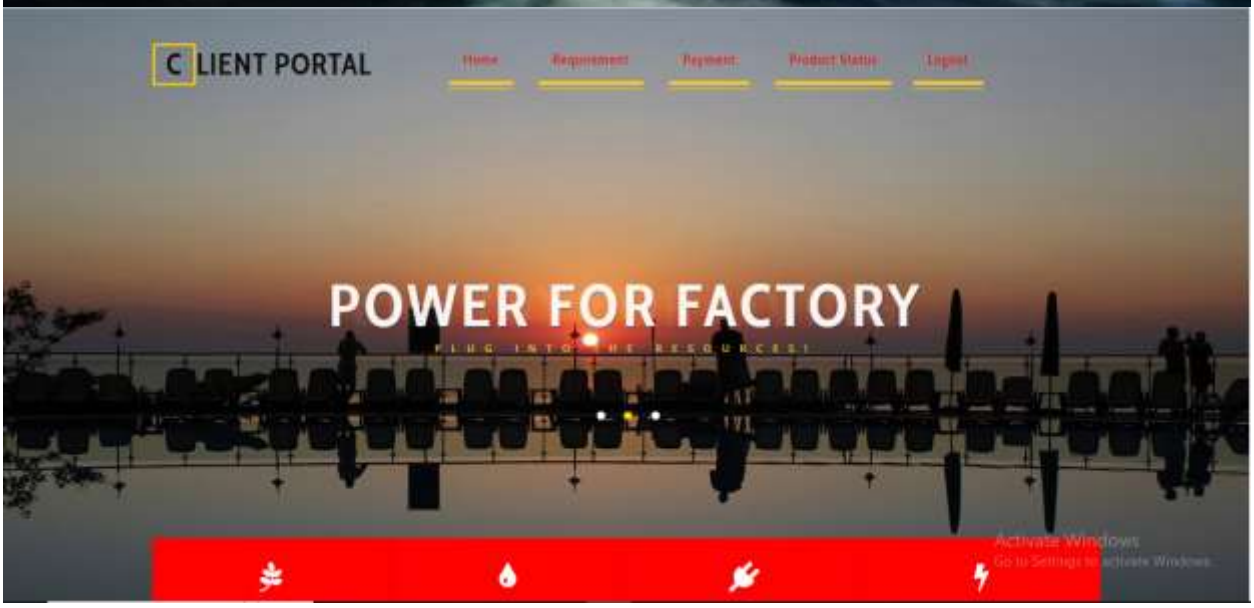


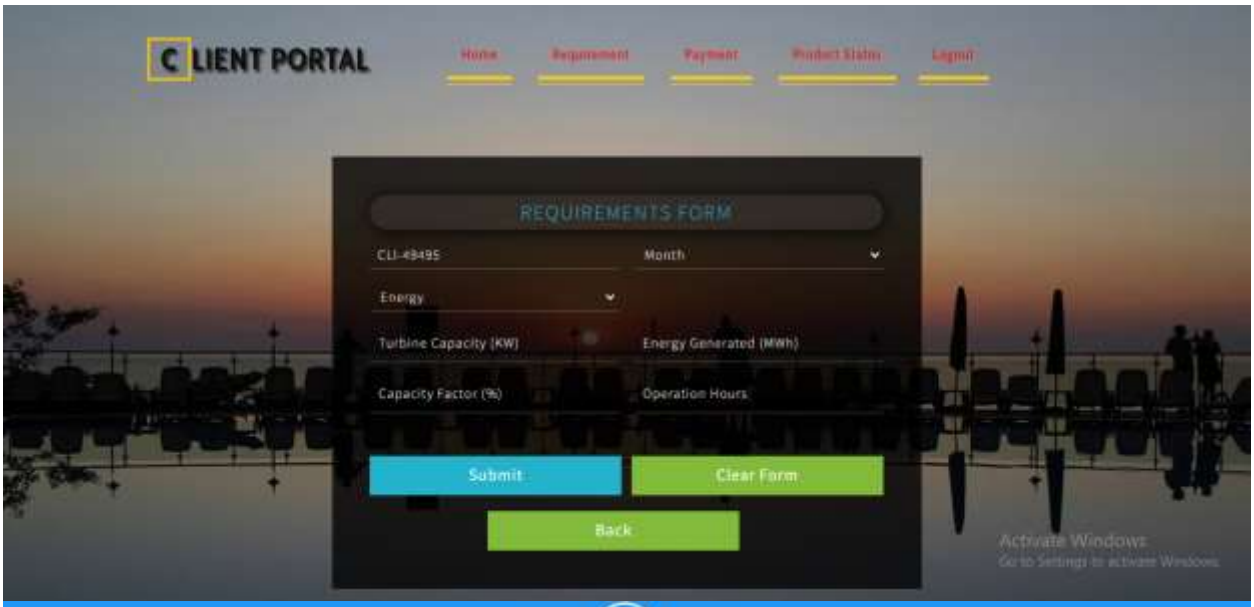
CHAPTER-7

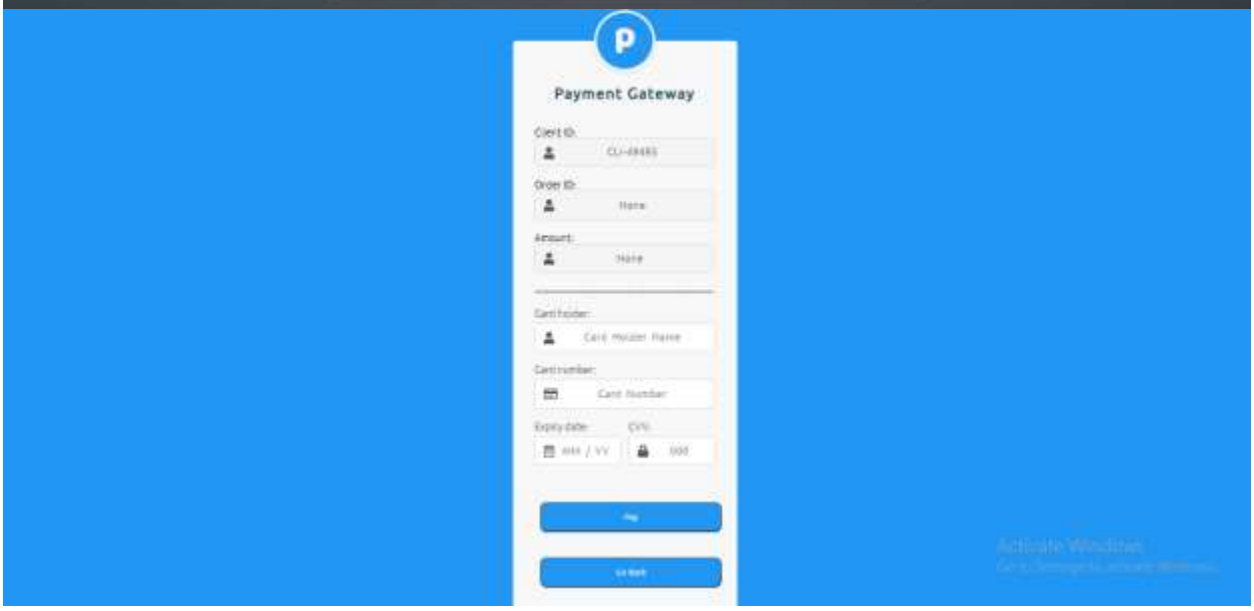
OUTPUT SCREENS

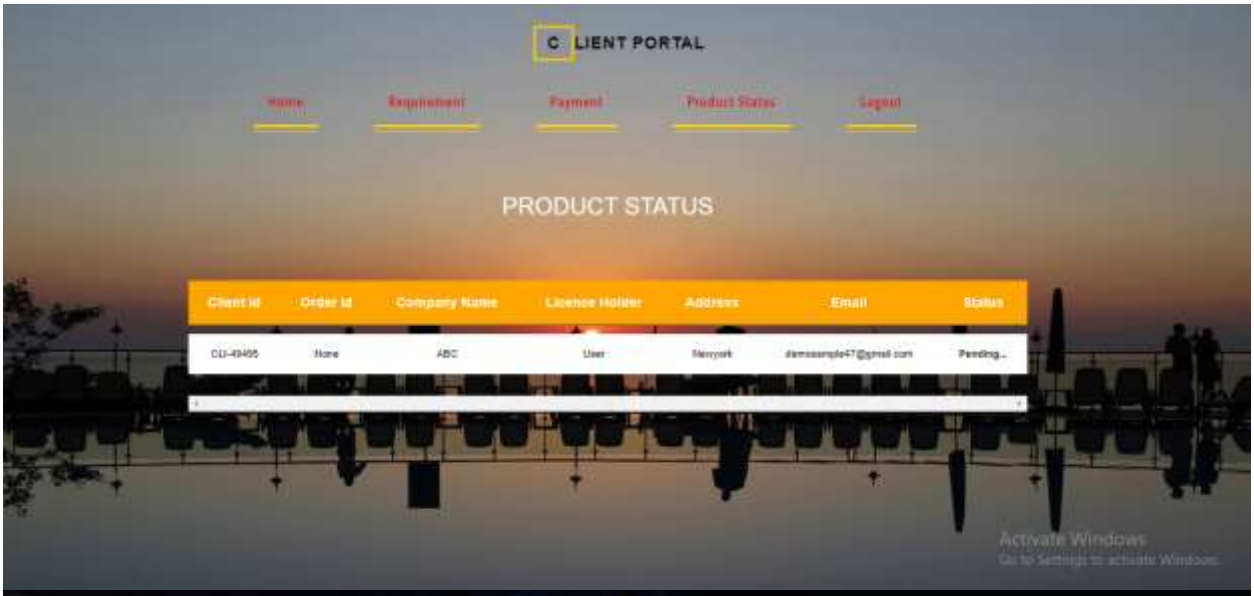

GRAVITY WELL
Client
Power Source
Grid Structure
System Maintenance
Admin
We Execute Our Ideas
From Start to Finish
"Energy Vault: Raising the bar in sustainable energy storage, one block at a time."
REGISTER FORM
REGISTER NOW
Company Name
Business Operator' Name
Company Address
Email
REGISTER
Already have an account, Click Login
LOGIN


LOGIN FORM
LOGIN NOW
Email
Password
LOGIN


CLIENT PORTAL
Home
Requirement
Payment
Product Status
Logout
POWER FOR FACTORY
Activate Windows


CLIENT PORTAL
Home
Requirement
Payment
Product Status
Logout
REQUIREMENTS FORM
CLI-49495
Month
Energy
Turbine Capacity (KW)
Energy Generated (MWh)
Capacity Factor (%)
Operation Hours
Submit
Clear Form
Back
Activate Windows
Go to Settings to activate Windows.


Payment Gateway
Client ID:
CLI-49495
Order ID:
None
Amount:
None
Cardholder:
Card Holder Name
Card number:
Card Number
Expiry date:
MM / YY
CVV:
000
Pay
Go Back


CLIENT PORTAL
Home
Requirement
Payment
Product Status
Logout
PRODUCT STATUS
Client Id
Order Id
Company Name
License Holder
Address
Email
Status
CLI-49495
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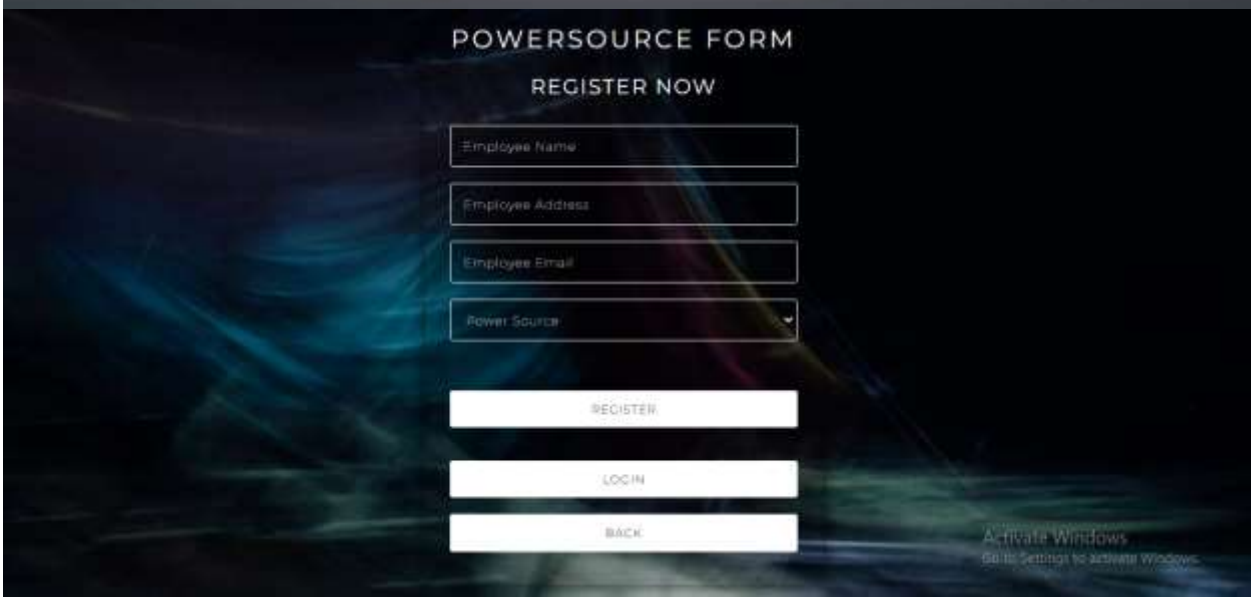
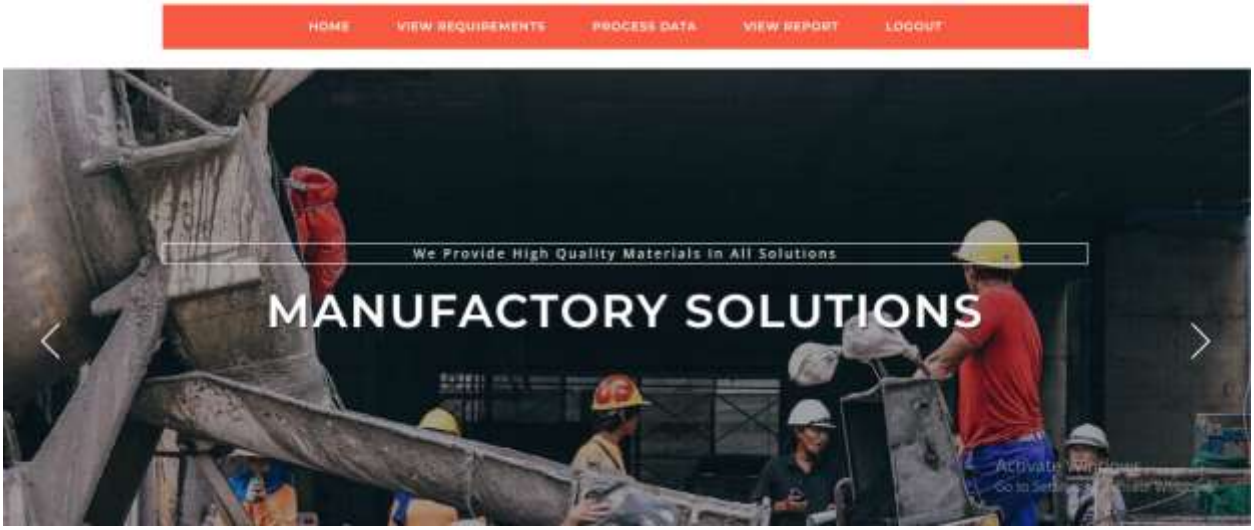
ABC
User
Newyork
demosample47@gmail.com
Pending...
Activate Windows
Go to Settings to activate Windows.


POWERSOURCE FORM
REGISTER NOW
Employee Name
Employee Address
Employee Email
Power Source
REGISTER
LOGIN
BACK
Activate Windows
Go to Settings to activate Windows.

POWER SOURCE



POWER SOURCE

HOME VIEW REQUIREMENTS PROCESS DATA VIEW REPORT LOGOUT

REQUIREMENTS DETAILS

Client Id	Month	Energy	Turbine Capacity	energy Generation	Capacity Factor	Operation Hours
CLI-49495	September	Wind Energy	2500.0	6.0	20.0	120

Activate Windows
Go to Settings to activate Windows.

POWER SOURCE

HOME VIEW REQUIREMENTS PROCESS DATA VIEW REPORT LOGOUT

POWERSOURCE PROGRESS

Client Id	Month	Energy	Turbine Capacity	Energy Generation	Capacity Factor	Operation Hours	Operation Hours
CLI-49495	September	Wind Energy	2500.0	6.0	20.0	120	PROGRESS

Activate Windows
Go to Settings to activate Windows.

POWER SOURCE

HOME VIEW REQUIREMENTS PROCESS DATA VIEW REPORT LOGOUT

POWERSOURCE REPORT

Client Id	Month	Energy	Gravity Potential Energy	Energy Turbine Capacity	Energy Generation Hr	Turbine Area
CLI-49495	September	Wind Energy	3049.66	1.43	249.23	182.03

Activate Windows
Go to Settings to activate Windows.

GRID STRUCTURE FORM

REGISTER NOW

Employee Name

Employee Address

Employee Email

Grid Structure

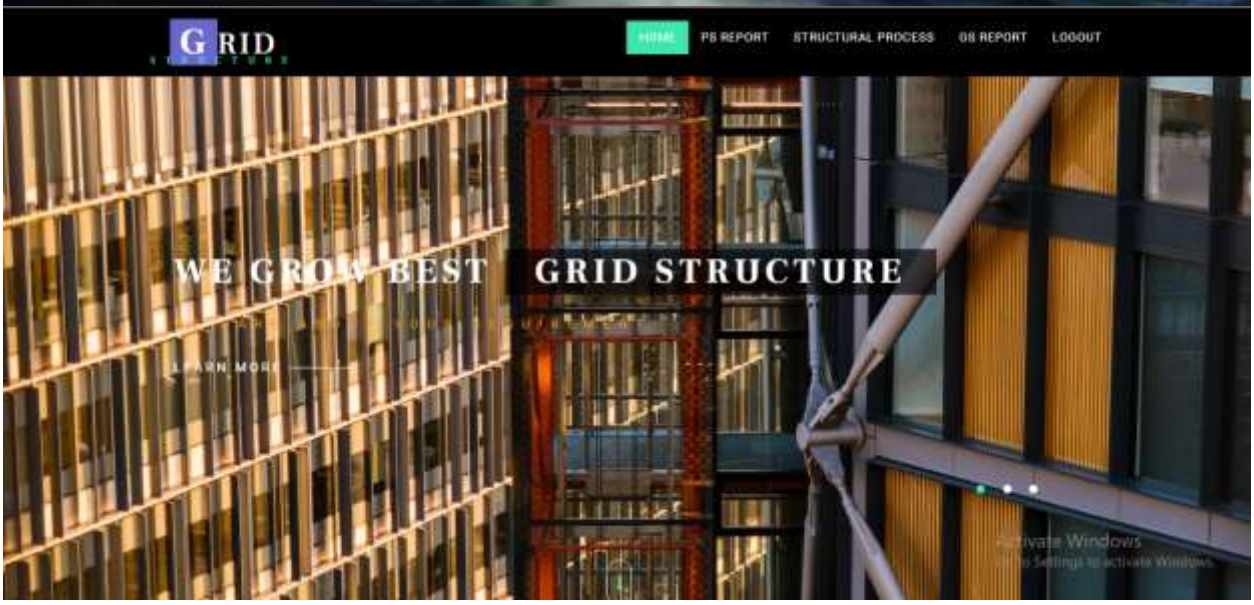
REGISTER

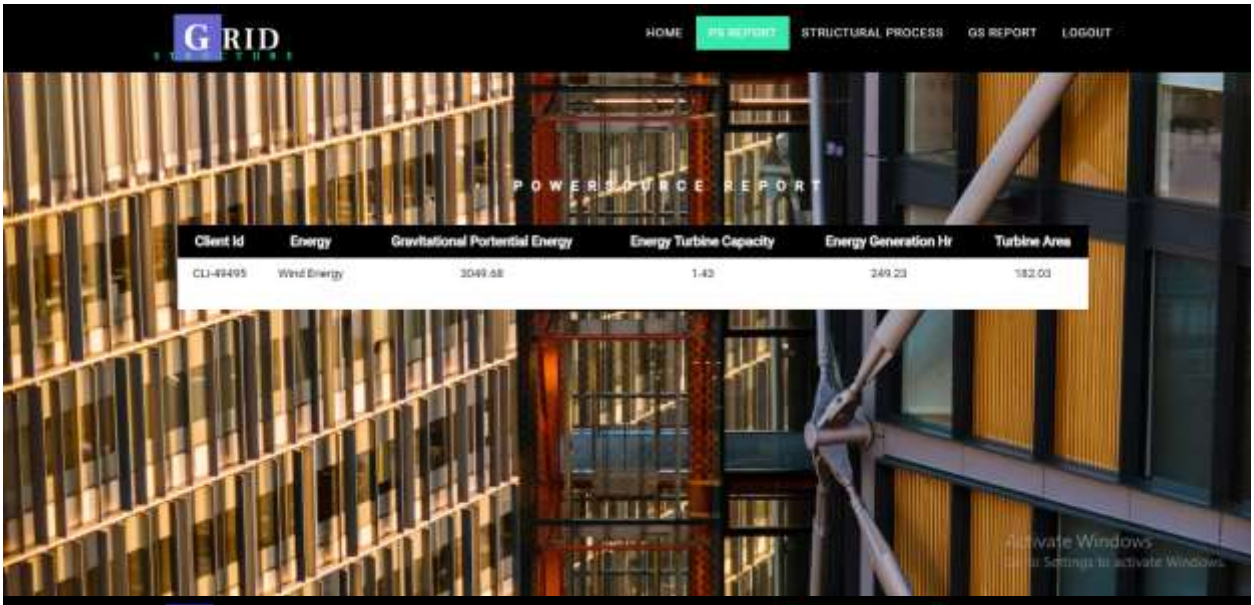
LOGIN

BACK

Activate Windows
Go to Settings to activate Windows.

GRID STRUCTURE LOGIN
LOGIN NOW
Email
Password
LOGIN
BACK
Activate Windows


GRID
HOME
PS REPORT
STRUCTURAL PROCESS
GS REPORT
LOGOUT
WE GROW BEST
GRID STRUCTURE
LEARN MORE


GRID
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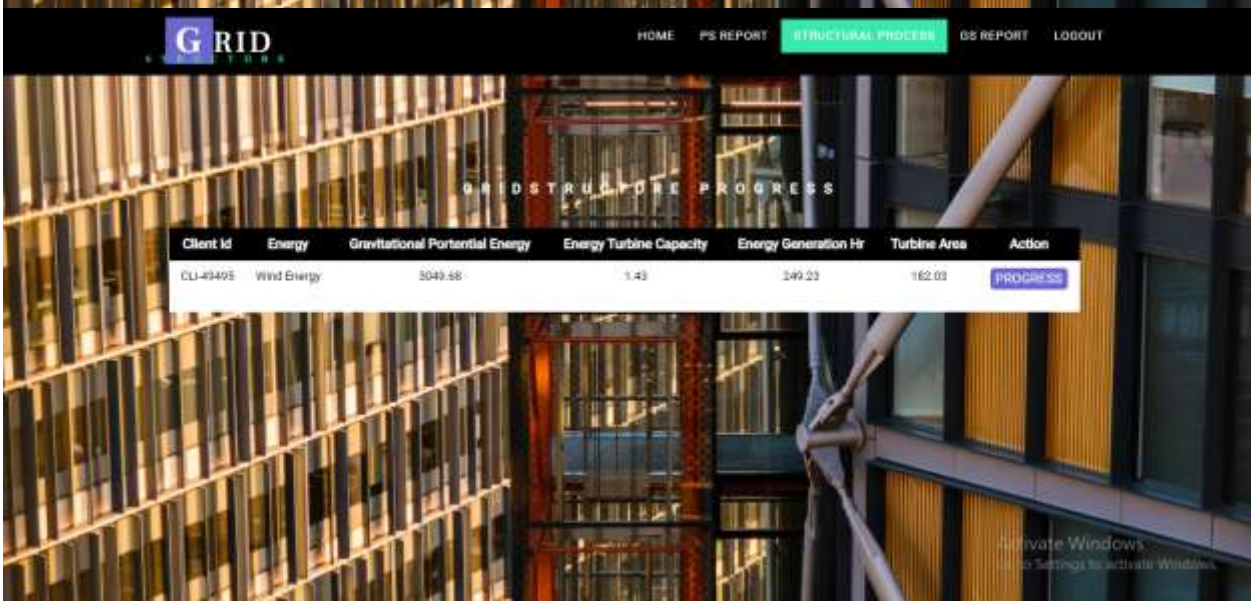
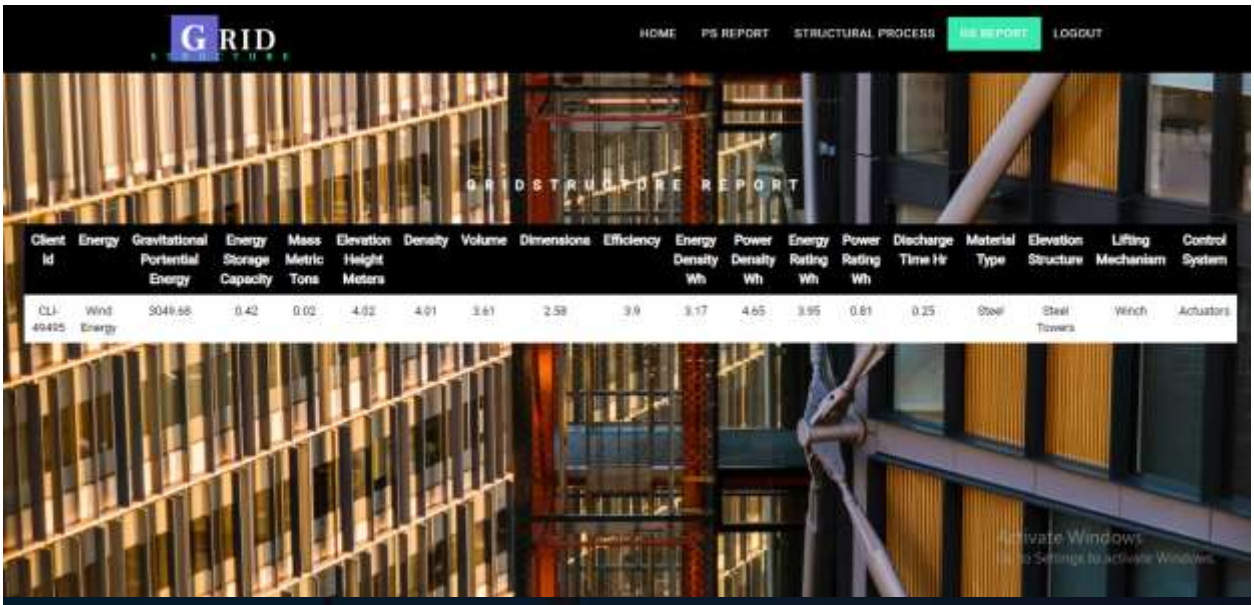
PS REPORT
STRUCTURAL PROCESS
GS REPORT
LOGOUT
POWERSOURCE REPORT
Client Id
Energy
Gravitational Portential Energy
Energy Turbine Capacity
Energy Generation Hr
Turbine Area
CLI-49495
Wind Energy
3049.68
1.43
249.23
182.03


GRID
HOME
PS REPORT
STRUCTURAL PROCESS
GS REPORT
LOGOUT
GRIDSTRUCTURE PROGRESS
Client Id
Energy
Gravitational Portential Energy
Energy Turbine Capacity
Energy Generation Hr
Turbine Area
Action
CLI-49495
Wind Energy
3049.68
1.43
249.23
182.03
PROGRESS



Client Id	Energy	Gravitational Portential Energy	Energy Storage Capacity	Mass Metric Tons	Elevation Height Meters	Density	Volume	Dimensions	Efficiency	Energy Density Wh	Power Density Wh	Energy Rating Wh	Power Rating Wh	Discharge Time Hr	Material Type	Elevation Structure	Lifting Mechanism	Control System
CLI-49495	Wind Energy	9049.68	0.42	0.02	4.02	4.01	3.61	2.58	3.9	3.17	4.65	3.95	0.81	0.25	Steel	Steel Towers	Winch	Actuators

SYSTEM MAINTENANCE

Home | GridStructure Report | Process System Maintenance | System Maintenance Report | Logout

GRIDSTRUCTURE REPORT

Client Id	Energy	Gravitational Portential Energy	Energy Storage Capacity	Mass Metric Tons	Elevation Height Meters	Density	Volume	Dimensions	Efficiency	Energy Density Wh	Power Density Wh	Energy Rating Wh	Power Rating Wh	Discharge Time Hr
CLI-49495	Wind Energy	3049.68	0.42	0.02	4.02	4.01	3.61	2.58	3.9	3.17	4.65	3.95	0.81	0.25

Activate Windows
Go to Settings to activate Windows.

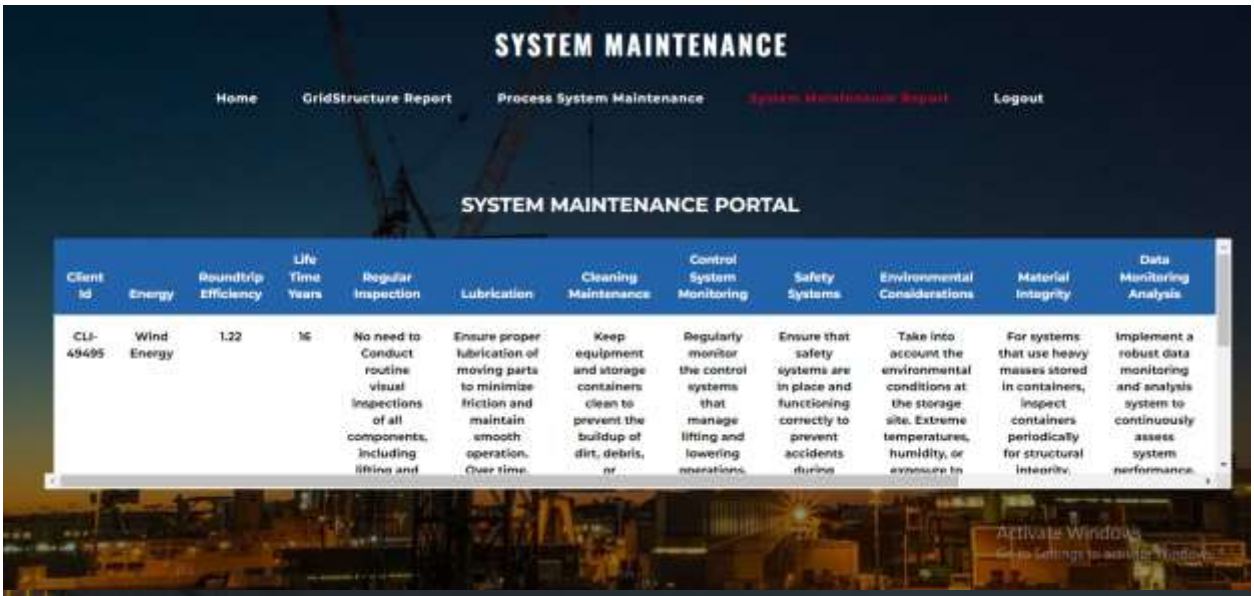
SYSTEM MAINTENANCE

Home | GridStructure Report | Process System Maintenance | System Maintenance Report | Logout

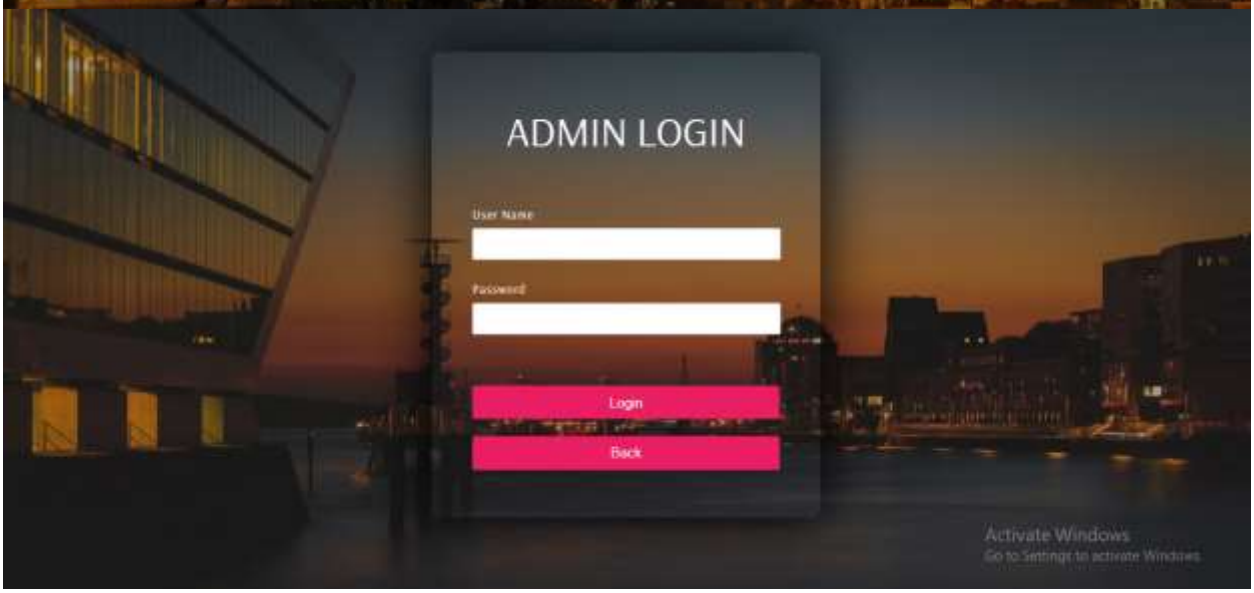
SYSTEM MAINTENANCE PORTAL

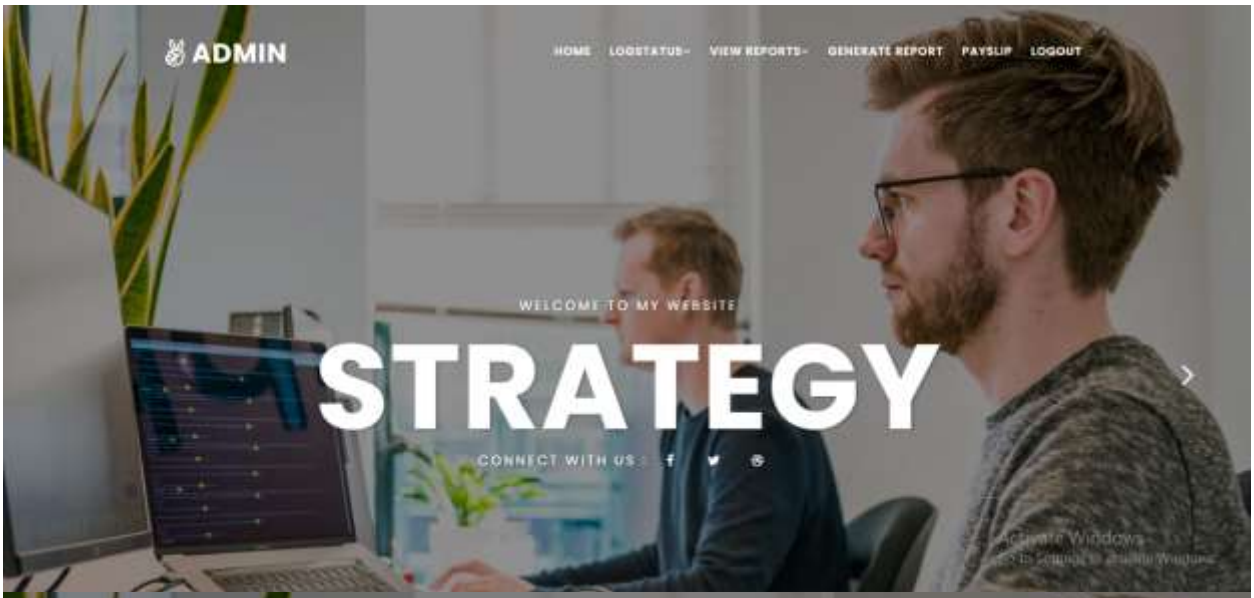
Density	Volume	Dimensions	Efficiency	Energy Density Wh	Power Density Wh	Energy Rating Wh	Power Rating Wh	Discharge Time Hr	Material Type	Elevation Structure	Lifting Mechanism	Control System	Action
4.01	3.61	2.58	3.9	3.17	4.65	3.95	0.81	0.25	Steel	Steel Towers	Winch	Actuators	PROCESS

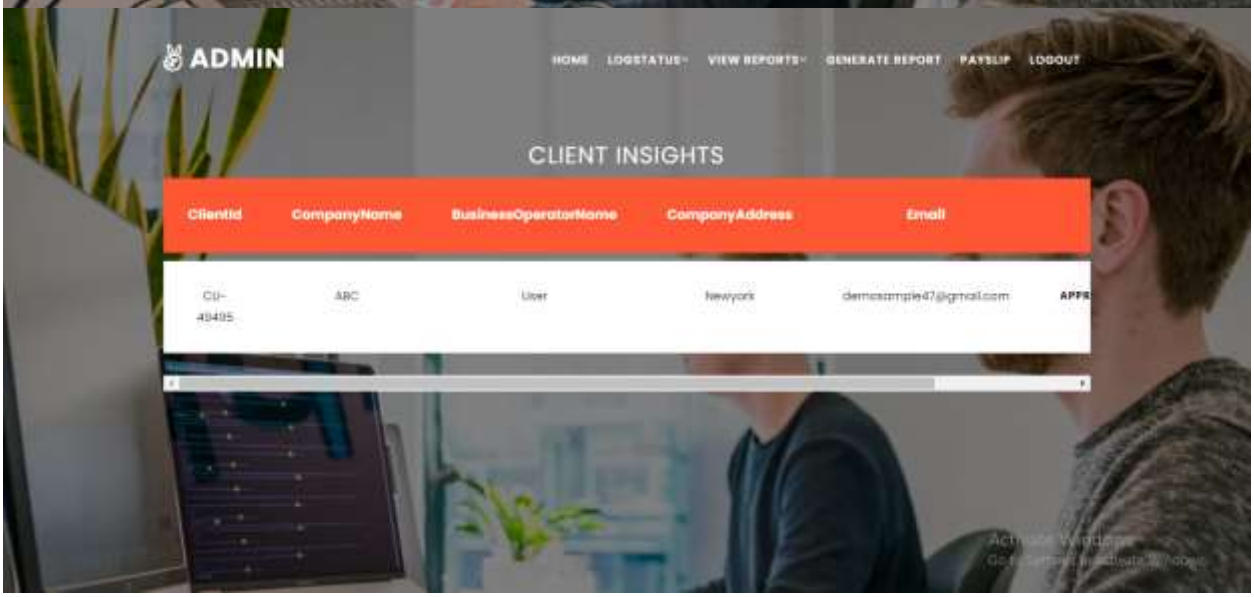
Activate Windows
Go to Settings to activate Windows.

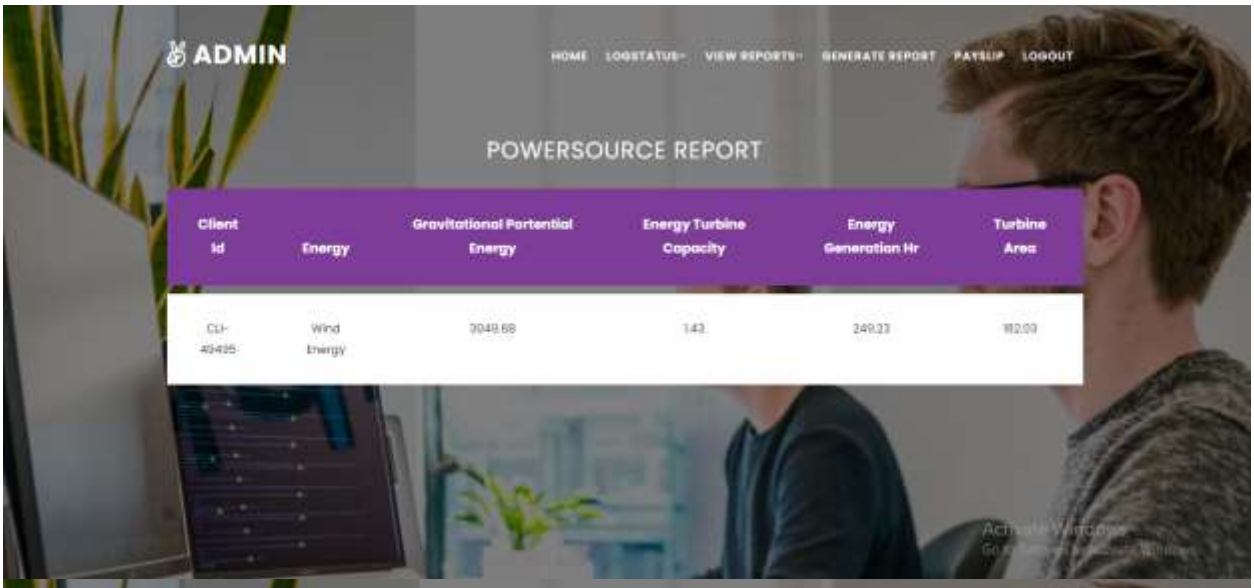


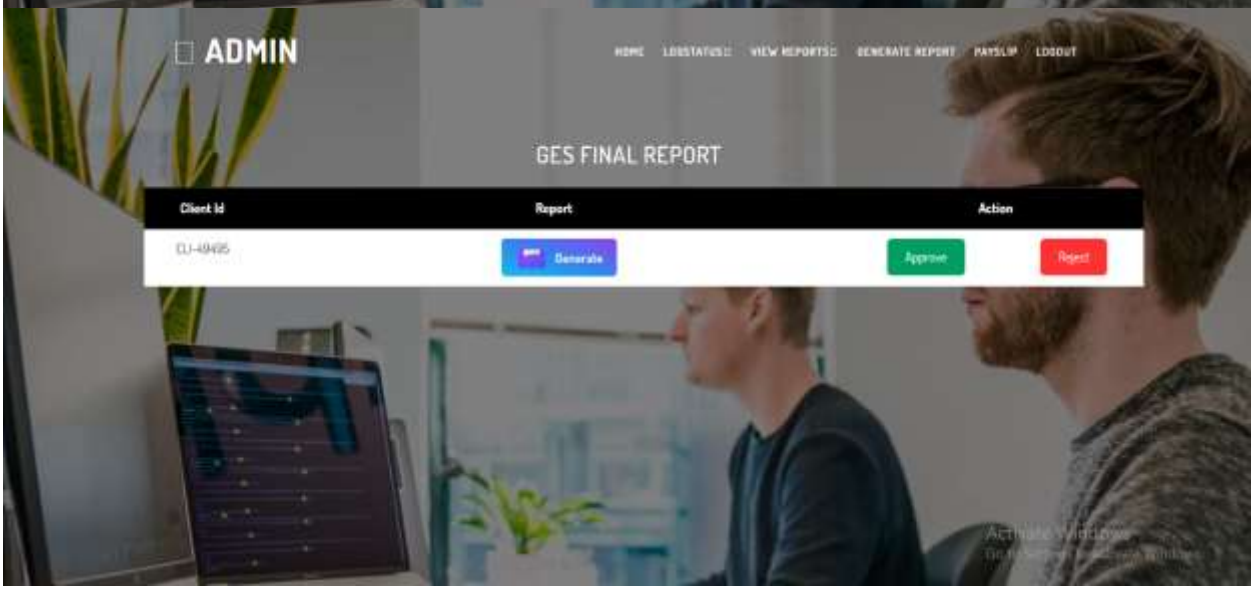
Client Id	Energy	Roundtrip Efficiency	Life Time Years	Regular Inspection	Lubrication	Cleaning Maintenance	Control System Monitoring	Safety Systems	Environmental Considerations	Material Integrity	Data Monitoring Analysis
CLI-49495	Wind Energy	1.22	16	No need to Conduct routine visual inspections of all components, including lifting and	Ensure proper lubrication of moving parts to minimize friction and maintain smooth operation. Over time	Keep equipment and storage containers clean to prevent the buildup of dirt, debris, or	Regularly monitor the control systems that manage lifting and lowering operations.	Ensure that safety systems are in place and functioning correctly to prevent accidents during	Take into account the environmental conditions at the storage site. Extreme temperatures, humidity, or exposure to	For systems that use heavy masses stored in containers, inspect containers periodically for structural integrity.	Implement a robust data monitoring and analysis system to continuously assess system performance.

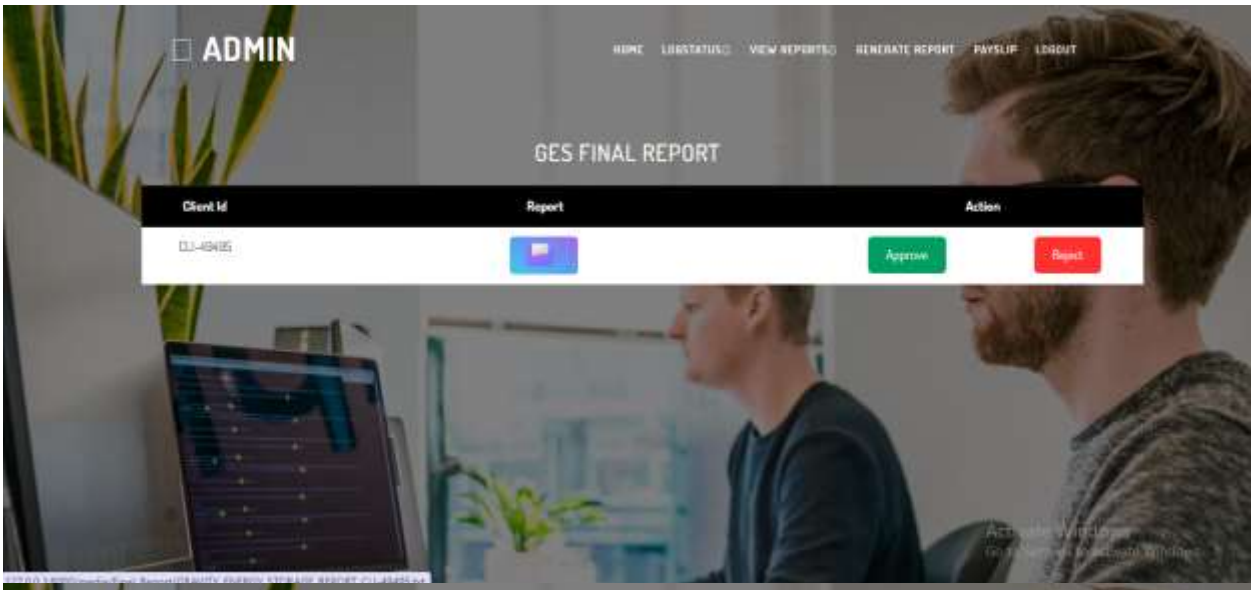


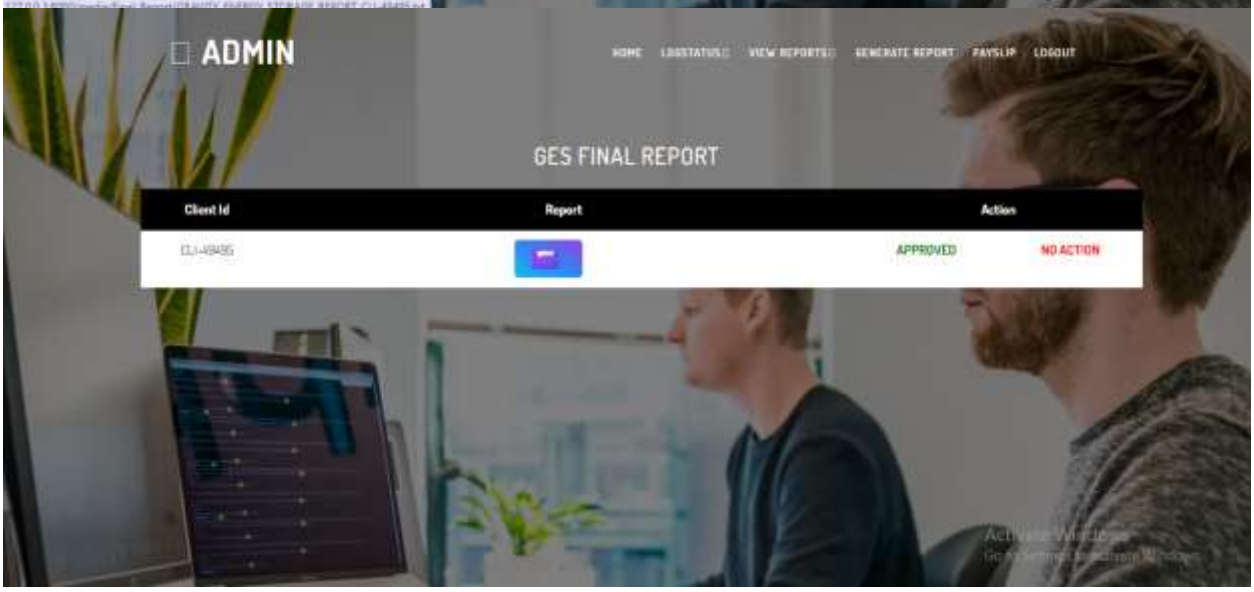

ADMIN
HOME
LOGSTATUS
VIEW REPORTS
GENERATE REPORT
PAYSLIP
LOGOUT
WELCOME TO MY WEBSITE
STRATEGY
CONNECT WITH US :


ADMIN
HOME
LOGSTATUS
VIEW REPORTS
GENERATE REPORT
PAYSLIP
LOGOUT
CLIENT INSIGHTS
ClientId
CompanyName
BusinessOperatorName
CompanyAddress
Email
CU-49495
ABC
User
Newyork
demosample47@gmail.com

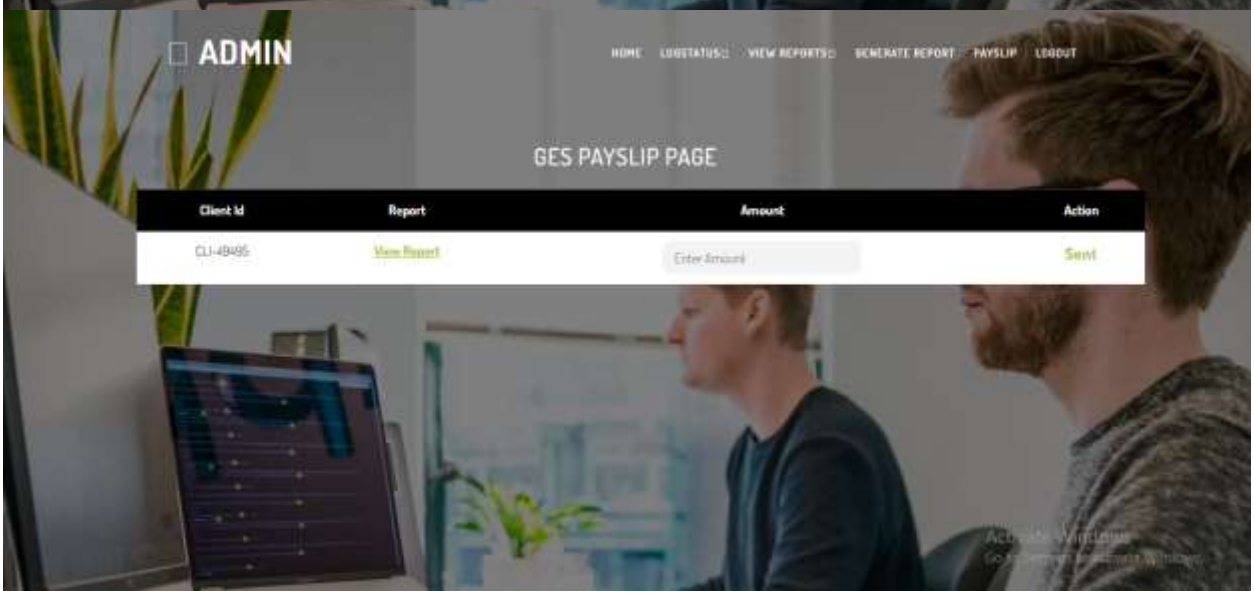

ADMIN
HOME
LOGSTATUS
VIEW REPORTS
GENERATE REPORT
PAYSLIP
LOGOUT
POWERSOURCE REPORT
Client Id
Energy
Gravitational Portential Energy
Energy Turbine Capacity
Energy Generation Hr
Turbine Area
CU-49495
Wind Energy
3048.68
143
249.23
182.03

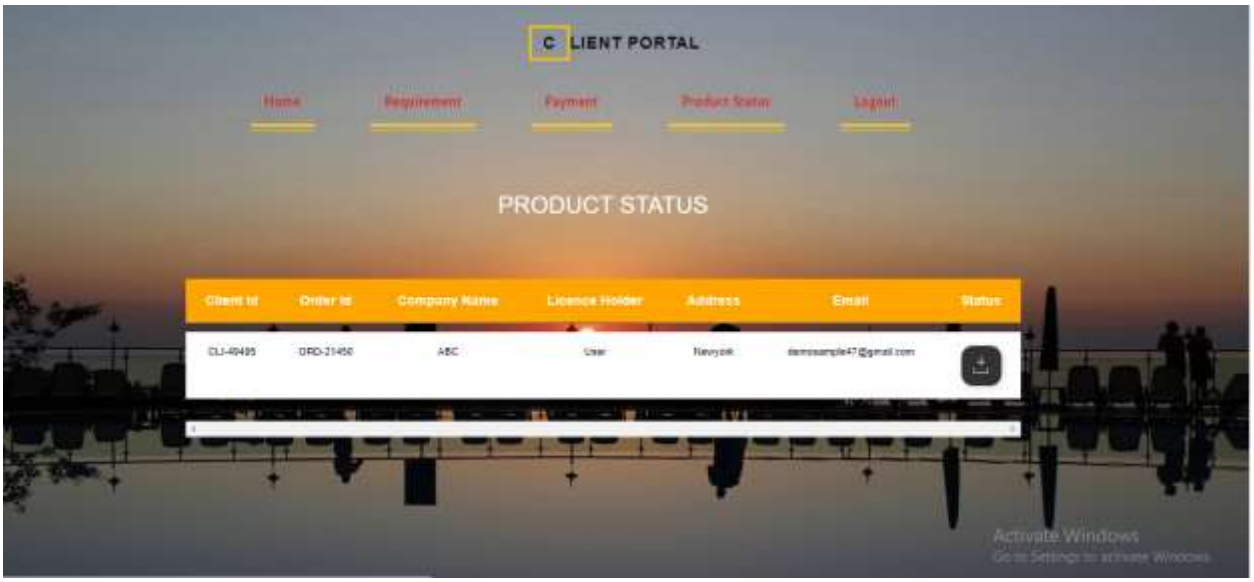

ADMIN
HOME
LOGSTATUS
VIEW REPORTS
GENERATE REPORT
PAYSLIP
LOGOUT
GES FINAL REPORT
Client Id
Report
Action
CU-49495
Generate
Approve
Reject


ADMIN
HOME
VIEW REPORTS
GENERATE REPORT
PAYSLIP
LOGOUT
GES FINAL REPORT
Client Id
Report
Action
Approve
Reject


ADMIN
HOME
VIEW REPORTS
GENERATE REPORT
PAYSLIP
LOGOUT
GES FINAL REPORT
Client Id
Report
Action
APPROVED
NO ACTION

ADMIN
HOME
LOGSTATUS
VIEW REPORTS
GENERATE REPORT
PAYSLIP
LOGOUT
GES PAYSLIP PAGE
Client Id
Report
Amount
Action
CLI-48495
View Report
Enter Amount
SEND PAYSLIP


ADMIN
HOME
LOGSTATUS
VIEW REPORTS
GENERATE REPORT
PAYSLIP
LOGOUT
GES PAYSLIP PAGE
Client Id
Report
Amount
Action
CLI-48495
View Report
Enter Amount
Sent


CLIENT PORTAL
Home
Requirement
Payment
Product Status
Logout
PRODUCT STATUS
Client Id
Order Id
Company Name
Licence Holder
Address
Email
Status
CLI-49495
ORD-21450
ABC
User
Newyork
demosample47@gmail.com
Activate Windows
Go to Settings to activate Windows.

CHAPTER-8

CODINGS

```
<!doctype html>
{% load static %}
<html lang="en">

<head>
    <!-- Required meta tags -->
    <meta charset="utf-8">
    <meta name="viewport" content="width=device-width, initial-scale=1,
shrink-to-fit=no">
    <title>Execution - Corporate Category Bootstrap Responsive Website
Template - Home : W3Layouts</title>
    <!-- google-fonts -->
    <link
href="//fonts.googleapis.com/css2?family=Montserrat:wght@100;200;300;400;500;
600;700;800;900&display=swap"
        rel="stylesheet">
    <!-- //google-fonts -->
    <!-- Font-Awesome-Icons-CSS -->
    <link rel="Shortcut icon" type="image/x-icon" href="{% static
'icons/home.jpg' %}" />
    <link rel="stylesheet" href="{% static 'home/css/fontawesome-all.min.css'
%}">
    <!-- Template CSS Style link -->
    <link rel="stylesheet" href="{% static 'home/css/style-starter.css' %}">
</head>

<body>
<script>
{% for msg in messages %}
    alert('{{ msg }}');
{% endfor %}
</script>
    <!--header-->
    <header id="site-header" class="fixed-top">
        <div class="container">
            <nav class="navbar navbar-expand-lg stroke">
                <h1>
                    <a class="navbar-brand d-flex align-items-center"
href="#">
                        <span class="lostyle-1">G</span>
                        ravity Well
                    </a>
                </h1>
                <!-- if logo is image enable this
    <a class="navbar-brand" href="#admin login.html">
        <img src="image-path" alt="Your logo" title="Your logo"
style="height:35px;" />
    </a> -->
                <button class="navbar-toggler  collapsed bg-gradient"
type="button" data-toggle="collapse"
                    data-target="#navbarTogglerDemo02" aria-
controls="navbarTogglerDemo02" aria-expanded="false"
                    aria-label="Toggle navigation">
                    <span class="navbar-toggler-icon fa icon-expand fa-
bars"></span>
                    <span class="navbar-toggler-icon fa icon-close fa-
```

```
times"></span>
                </button>

                <div class="collapse navbar-collapse"
id="navbarTogglerDemo02">
                    <ul class="navbar-nav ml-lg-auto">

                         <li class="nav-item active">
                            <a class="nav-link" href="/">HOME<span class="sr-
only">(current)</span></a>
                        </li><li class="nav-item">
                            <a class="nav-link" href="/register/">CLIENT<span
class="sr-only">(current)</span></a>
                        </li>
                        <li class="nav-item">
                            <a class="nav-link" href="/registeration/">POWER
SOURCE</a>
                        </li>
                        <li class="nav-item">
                            <a class="nav-link" href="/gregister/">GRID
STRUCTURE</a>
                        </li>
                        <li class="nav-item">
                            <a class="nav-link" href="/sregister/">SYSTEM
MAINTENANCE</a>
                        </li>
                         <li class="nav-item">
                            <a class="nav-link"
href="/manager_login/">ADMIN</a>
                        </li>

                    </ul>
                </div>

                <div class="cont-ser-position">
                    <nav class="navigation">
                        <div class="theme-switch-wrapper">
                            <label class="theme-switch" for="checkbox">
                                <input type="checkbox" id="checkbox">
                                <div class="mode-container">
                                    <i class="gg-sun"></i>
                                    <i class="gg-moon"></i>
                                </div>
                            </label>
                        </div>
                    </nav>
                </div>
                <!-- //toggle switch for light and dark theme -->
            </nav>
        </div>
    </header>
    <!--//header-->

    <!-- banner section -->
    <section class="banner-19">
        <div class="banner-layer">
            <video autoplay muted loop id="myVideo">
```

```
                <source src="{% static 'home/videos/video.mp4' %}"
type="video/mp4">
                Your browser does not support HTML5 video.
            </video>
            <div class="main-content-top">
                <div class="container">
                    <div class="main-content">
                        <!-- if logo is image enable this
                        <a class="logo" href="admin_login.html">
                            <img src="image-path" alt="Your logo" title="Your
logo" style="height:35px;" />
                        </a> -->
                        <h4>We Execute Our Ideas From Start to Finish </h4>
                        <p class="mt-3">
"Energy Source: Raising the bar in sustainable energy storage, one block at a
time."</p>

                    </div>
                </div>
            </div>
        </div>
    </section>

    <section class="w3l-aboutblock py-5">
        <div class="container py-md-5 py-sm-4">
            <div class="row">
                <div class="col-lg-6 left-wthree-img mb-lg-0 mb-4">
                    <img class="img-fluid img-responsive" src="{% static
'home/images/about.jpg' %}" alt=" ">
                </div>
                <div class="col-lg-6 about-right-faq align-self pl-lg-5">
                    <h3 class="title-style">The Largest Business Expert!</h3>
                    <p class="mt-2">The Gravity Storage System stands as the
epitome of business expertise, representing the forefront of innovation and
efficiency in energy storage.</p>
                      <p class="mt-2"> This revolutionary system harnesses
the power of gravity to store and release energy, offering unprecedented
scalability and reliability.</p>
                     <p class="mt-2">  As the largest and most advanced
business expert in energy storage, the Gravity Storage System ensures
seamless integration with diverse industries,
                     providing a sustainable and cost-effective solution for
meeting the ever-growing demands of modern businesses. </p>
</div>
            </div>
        </div>
    </section>
    <div class="service-section py-5">
        <div class="container py-md-5 py-4">
            <div class="row align-items-center mb-5">
                <div class="col-md-6 section-heading">
<!--                    <h5 class="small-title-2">Web Services</h5>-->
                    <h3 class="title-style-2">
                        Our Best Services</h3>
                </div>
                <div class="col-md-6 section-heading mt-md-0 mt-2">
                    <p>
```

```
                        Our best services encompass cutting-edge energy
storage solutions and unparalleled expertise in optimizing power consumption,
providing clients with innovative and sustainable solutions for their
evolving energy needs.
                    </p>
                </div>
            </div>
            <div class="row justify-content-center">
                <div class="col-lg-4 col-md-6 item">
                    <div class="card">
                        <div class="card-header p-0 position-relative border-
0">
                            <a href="services.html">
                                <img class="d-block img-responsive" src="{%
static 'home/images/s4.jpg' %}" alt="card-image">
                            </a>
                        </div>
                        <div class="card-body service-details">
                            <span class="label-style">Work</span>
                            <a href="services.html" class="service-
heading"></a>
                            <p>Under development, the Gravity Storage System
promises to be the largest business expert in energy storage, leveraging
gravity to redefine efficiency and scalability. Stay tuned for a
groundbreaking solution that will revolutionize how businesses approach
energy storage challenges.</p>
                        </div>
                    </div>
                </div>
                <div class="col-lg-4 col-md-6 mt-md-0 mt-4">
                    <div class="card">
                        <div class="card-header p-0 position-relative border-
0">
                            <a href="services.html">
                                <img class="d-block img-responsive" src="{%
static 'home/images/s5.jpg' %}" alt="card-image">
                            </a>
                        </div>
                        <div class="card-body service-details">
                            <span class="label-style label-style-
2">Business</span>
                            <a href="services.html" class="service-heading">
</a>
                            <p>In the world of business, the Gravity Storage
System reigns supreme, offering unparalleled energy storage capabilities
through innovative gravity-based technology. Its scalable and reliable design
positions it as the foremost choice for businesses seeking efficient and
sustainable solutions.</p>
                        </div>
                    </div>
                </div>
                <div class="col-lg-4 col-md-6 mt-lg-0 mt-4">
                    <div class="card">
                        <div class="card-header p-0 position-relative border-
0">
                            <a href="services.html">
                                <img class="d-block img-responsive" src="{%
```

```
static 'home/images/s2.jpg' %}" alt="card-image">
                            </a>
                        </div>
                        <div class="card-body service-details">
                            <span class="label-style label-style-
3">Corporate</span>
                            <a href="services.html" class="service-heading">
</a>
                            <p>Elevate your corporate operations with the
Gravity Storage System — a cutting-edge energy storage solution that
optimizes efficiency and sustainability,
                                setting new benchmarks for corporate
excellence. Revolutionize your energy strategy with the largest business
expert in the field of Gravity Storage System.
                            </p>
                        </div>
                    </div>
                </div>
            </div>
        </div>
    </div>
    <!-- //services section -->

    <!-- about section -->
    <div class="w3l-content-photo-5 py-5">
        <div class="container py-lg-5 py-sm-4">
            <div class="row align-items-center">
                <div class="col-lg-6 content-image order-lg-first order-
last">
                    <a href="#image"><img src="{% static
'home/images/img2.jpg' %}" class="img-responsive" alt="content-photo"></a>
                </div>
                <div class="col-lg-6 col-md-10 mx-auto content-right mb-lg-0
mb-5 pl-lg-5 order-lg-last order-first">
                    <h5 class="small-title-2">About the Company</h5>
                    <h3 class="title-style-2">Creative Solutions</h3>
                    <p class="my-3">Gravity energy storage, like pumped
storage hydropower, efficiently stores and releases electricity using
gravitational potential energy. With two reservoirs at different elevations,
surplus energy pumps water uphill, converting it to gravitational potential
energy. </p>
                    <p class="border-top-w3l pt-4 mt-4">During peak demand,
the stored energy is released downhill, powering turbines for electricity
generation. Renowned for reliability and scalability, gravity energy storage
provides significant capacities and rapid response, playing a crucial role in
grid stabilization to balance power supply and demand.</p>


    <script>
        // When the user scrolls down 20px from the top of the document, show
the button
        window.onscroll = function () {
            scrollFunction()
        };

        function scrollFunction() {
            if (document.body.scrollTop > 20 ||
```

```
document.documentElement.scrollTop > 20) {
                document.getElementById("movetop").style.display = "block";
            } else {
                document.getElementById("movetop").style.display = "none";
            }
        }

        // When the user clicks on the button, scroll to the top of the
document
        function topFunction() {
            document.body.scrollTop = 0;
            document.documentElement.scrollTop = 0;
        }
    </script>
    <!-- //move top -->

    <!-- common jquery plugin -->
    <script src="{% static 'home/js/jquery-3.3.1.min.js' %}"></script>
    <!-- //common jquery plugin -->

    <!-- owl carousel -->
    <script src="{% static 'home/js/owl.carousel.js' %}"></script>
    <!-- script for tesimonials carousel slider -->
    <script>
        $(document).ready(function () {
            $("#owl-demo2").owlCarousel({
                loop: true,
                nav: false,
                margin: 50,
                responsiveClass: true,
                autoplay: true,
                autoplayTimeout: 5000,
                autoplaySpeed: 1000,
                autoplayHoverPause: true,
                responsive: {
                    0: {
                        items: 1,
                        nav: false
                    },
                    736: {
                        items: 1,
                        nav: false
                    },
                    991: {
                        items: 2,
                        margin: 30,
                        nav: false
                    },
                    1080: {
                        items: 3,
                        nav: false
                    }
                }
            })
        })
    </script>
    <!-- //script for tesimonials carousel slider -->
```

```
    <!-- owl carousel -->

    <!-- magnific popup -->
    <script src="{% static 'home/js/jquery.magnific-popup.min.js'
%}"></script>
    <script>
        $(document).ready(function () {
            $('.popup-with-zoom-anim').magnificPopup({
                type: 'inline',

                fixedContentPos: false,
                fixedBgPos: true,

                overflowY: 'auto',

                closeBtnInside: true,
                preloader: false,

                midClick: true,
                removalDelay: 300,
                mainClass: 'my-mfp-zoom-in'
            });

            $('.popup-with-move-anim').magnificPopup({
                type: 'inline',

                fixedContentPos: false,
                fixedBgPos: true,

                overflowY: 'auto',

                closeBtnInside: true,
                preloader: false,

                midClick: true,
                removalDelay: 300,
                mainClass: 'my-mfp-slide-bottom'
            });
        });
    </script>
    <!-- //magnific popup -->

    <!-- theme switch js (light and dark)-->
    <script src="{% static 'home/js/theme-change.js' %}"></script>
    <script>
        function autoType(elementClass, typingSpeed) {
            var thhis = $(elementClass);
            thhis.css({
                "position": "relative",
                "display": "inline-block"
            });
            thhis.prepend('<div class="cursor" style="right: initial;
left:0;"></div>');
            thhis = thhis.find(".text-js");
            var text = thhis.text().trim().split('');
            var amntOfChars = text.length;
            var newString = "";
```

```
            thhis.text("|");
            setTimeout(function () {
                thhis.css("opacity", 1);
                thhis.prev().removeAttr("style");
                thhis.text("");
                for (var i = 0; i < amntOfChars; i++) {
                    (function (i, char) {
                        setTimeout(function () {
                            newString += char;
                            thhis.text(newString);
                        }, i * typingSpeed);
                    })(i + 1, text[i]);
                }
            }, 1500);
        }

        $(document).ready(function () {
            // Now to start autoTyping just call the autoType function with
the
            // class of outer div
            // The second paramter is the speed between each letter is typed.
            autoType(".type-js", 200);
        });
    </script>
    <!-- //theme switch js (light and dark)-->

    <!-- MENU-JS -->
    <script>
        $(window).on("scroll", function () {
            var scroll = $(window).scrollTop();

            if (scroll >= 80) {
                $("#site-header").addClass("nav-fixed");
            } else {
                $("#site-header").removeClass("nav-fixed");
            }
        });

        //Main navigation Active Class Add Remove
        $(".navbar-toggler").on("click", function () {
            $("header").toggleClass("active");
        });
        $(document).on("ready", function () {
            if ($(window).width() > 991) {
                $("header").removeClass("active");
            }
            $(window).on("resize", function () {
                if ($(window).width() > 991) {
                    $("header").removeClass("active");
                }
            });
        });
    </script>
    <!-- //MENU-JS -->

    <!-- disable body scroll which navbar is in active -->
    <script>
```

```
        $(function () {
            $('.navbar-toggler').click(function () {
                $('body').toggleClass('noscroll');
            })
        });
    </script>
    <!-- //disable body scroll which navbar is in active -->

    <!--bootstrap-->
    <script src="{% static 'home/js/bootstrap.min.js' %}"></script>
    <!-- //bootstrap-->
    <!-- //Js scripts -->
</body>

</html>
```

SYSTEM TESTING AND IMPLEMENTATION

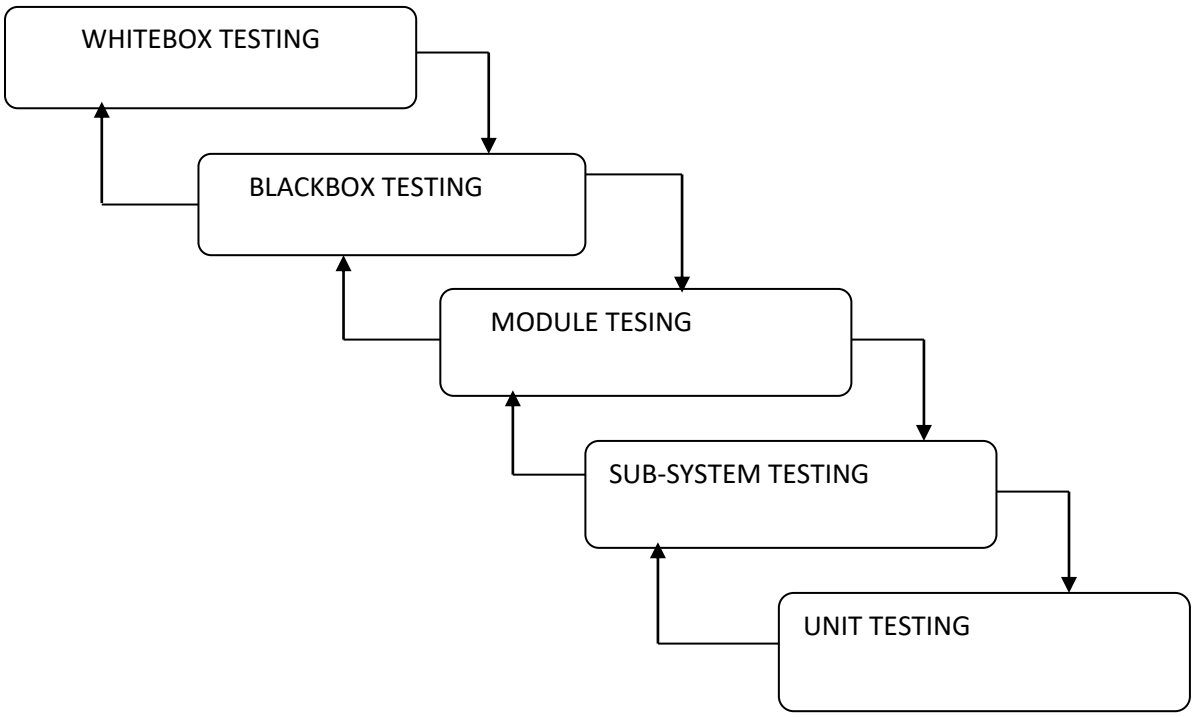
9.1. INTRODUCTION

Software testing is a critical element of software quality assurance and represents the ultimate review of specification, design and coding. In fact, testing is the one step in the software engineering process that could be viewed as destructive rather than constructive.

9.2. STRATEGIC APPROACH TO SOFTWARE TESTING

The software engineering process can be viewed as a spiral. Initially system engineering defines the role of software and leads to software requirement analysis where the information domain, functions, behavior, performance, constraints and validation criteria for software are established. Moving inward along the spiral, we come to design and finally to coding. To develop computer software we spiral in along streamlines that decrease the level of abstraction on each turn.

A strategy for software testing may also be viewed in the context of the spiral. Unit testing begins at the vertex of the spiral and concentrates on each unit of the software as implemented in source code. Testing progress is done by moving outward along the spiral to integration testing, where the focus is on the design and the construction of the software architecture. Talking another turn on outward on the spiral we encounter validation testing where requirements established as part of software requirements analysis are validated against the software that has been constructed. Finally, we arrive at system testing, where the software and other system elements are tested as a whole.



9.3. TESTING PROCEDURES

Unit testing focuses verification effort on the smallest unit of software design, the module. The unit testing we have is white box oriented and some modules the steps are conducted in parallel.

1. WHITE BOX TESTING

This type of testing ensures that

- All independent paths have been exercised at least once
- All logical decisions have been exercised on their true and false sides
- All loops are executed at their boundaries and within their operational bounds
- All internal data structures have been exercised to assure their validity.

To follow the concept of white box testing we have tested each form .We have created independently to verify that Data flow is correct, All conditions are exercised to check their validity, All loops are executed on their boundaries.

2. BLACK BOX TESTING

The established technique of flow graph with Cyclamate complexity was used to derive test cases for all the functions. The main steps in deriving test cases were:

Use the design of the code and draw correspondent flow graphs.

Determine the Cyclamate complexity of the resultant flow graph, using formula:

V (G) =E-N+2 or

V (G) =P+1 or

V (G) =Number of Regions

Where V (G) is Cyclomatic complexity,

E is the number of edges,

N is the number of flow graph nodes,

P is the number of predicate nodes.

Determine the basis of set of linearly independent paths.

3. MODULE TESTING

In this part of the testing each of the conditions were tested to both true and false aspects. And all the resulting paths were tested. So that each path that may be generated on particular condition is traced to uncover any possible errors.

4. SUB SYSTEM TESTING

This type of testing selects the path of the program, according to the location of the definition and use of variables. This kind of testing was used only when some local variable were declared. The definition-use chain method was used in this type of testing. These were particularly useful in nested statements.

5. UNIT TESTING

In this type of testing all the loops are tested to all the limits possible. The following exercise was adopted for all loops:

- All the loops were tested at their limits, just above them and just below them.
- All the loops were skipped at least once.
- For nested loop test the innermost loop first and then work outwards.
- For concatenated loops the values of dependent loops were set with the help of a connected loop.

CHAPTER-10

SYSTEM SECURITY

10.1 INTRODUCTION

Security system can be divided into four related issues: The protection of computer based resources that includes hardware, software, data, procedures and people against unauthorized use or natural

Disaster is known as System Security.

- Security
- Integrity
- Privacy
- Confidentiality

SYSTEM SECURITY refers to the technical innovations and procedures applied to the hardware and operation systems to protect against deliberate or accidental damage from a defined threat.

DATA SECURITY is the protection of data from loss, disclosure, modification and destruction.

SYSTEM INTEGRITY refers to the power functioning of hardware and programs, appropriate physical security and safety against external threats such as eavesdropping and wiretapping.

PRIVACY defines the rights of the user or organizations to determine what information they are willing to share with or accept from others and how the organization can be protected against unwelcome, unfair or excessive dissemination of information about it.

CONFIDENTIALITY is a special status given to sensitive information in a database to minimize the possible invasion of privacy. It is an attribute of information that characterizes its need for protection.

10.2 SECURITY IN SOFTWARE

System security refers to various validations on data in the form of checks and controls to avoid the system from failing. It is always important to ensure that only valid data is entered and only valid operations are performed on the system. The system employs two types of checks and controls:

CLIENT SIDE VALIDATION

Various client side validations are used to ensure on the client side that only valid data is entered. Client side validation saves server time and load to handle invalid data. Some checks are imposed:

- JavaScript in used to ensure those required fields are filled with suitable data only. Maximum lengths of the fields of the forms are appropriately defined.
- Forms cannot be submitted without filling up the mandatory data so that manual mistakes of submitting empty fields that are mandatory can be sorted out at the client side to save the server time and load.
- Tab-indexes are set according to the need and taking into account the ease of use while working with the system.

SERVER SIDE VALIDATION

Some checks cannot be applied on the client side. Server side checks are necessary to save the system from failing and intimating the user that some invalid operation has been performed or the performed operation is restricted. Some of the server side checks imposed is:

- A server side constraint has been imposed to check for the validity of primary key and foreign key. A primary key value cannot be duplicated. Any attempt to duplicate the primary value results in a message intimating the user about those values through the forms using foreign key can be updated only of the existing foreign key values.
- The user is intimated through appropriate messages about the successful operations or exceptions occurring at server side.
- Various Access Control Mechanisms have been built so that one user may not agitate upon another. Access permissions to various types of users are controlled according to the organizational structure. Only permitted users can log on to the system and can have access according to their category. User- name, passwords and permissions are controlled the server side.
- Using server side validation, constraints on several restricted operations are imposed.

CHAPTER-11

CONCLUSION

CONCLUSION:

In conclusion, The “Celestial Path Exploration through a Multifaceted Simulation System for Trajectory Analysis of Celestial Bodies” Project presents a transformative solution to the disjointed processes in current energy management systems. By introducing a unified platform, it streamlines processes, enhances transparency, and reduces manual dependencies. This user-centric and efficient approach ensures timely execution of tasks, optimizing energy infrastructure management and promoting the adoption of advanced methodologies. The future trajectory of the Gravity Project involves a commitment to continuous improvement and adaptability to emerging energy industry trends. The system will be designed for scalability, ensuring seamless integration of new features and functionalities as the energy landscape evolves. Strengthening security measures will be a priority to fortify the protection of sensitive energy data, guaranteeing the secure functioning of the platform. The project will also focus on user-centric strategies, implementing comprehensive training programs and support systems to empower all stakeholders in effectively utilizing the platform. Furthermore, collaboration with industry experts will be an ongoing initiative, enabling the system to remain dynamic and responsive to the evolving needs and challenges of energy infrastructure management, ensuring its continual relevance and effectiveness in the industry.